



MATHS in The Kitchen



Barbara Lynch and
Angelo Pantalone



Quality ready-to-use
resources

MATHS in The Kitchen



**Barbara Lynch and
Angelo Pantalone**

This book was commissioned by Jennifer Gough, project management was by Piper Print Media, Gail McManus Graphics designed the cover and David Dickson supplied the illustrations.

Barbara Lynch is former Head of Wesley College in Glen Waverley, Victoria, author of nearly 100 titles and now a consultant in areas such as strategic planning, governance, policy development, curriculum, assessment, reporting, CD-ROM and Internet publishing. Barbara has a long and enviable background in secondary education and mathematics.

Angelo Pantalone is well travelled and has a vast array of interests including cooking, politics and sociology. He is presently a lecturer at RMIT, teaching subjects about merchandising and product development in the fashion and textiles industry. As a part of the Merchandising Management Team at RMIT, Angelo has received numerous awards for teaching excellence. He also has experience in the fashion business through roles at companies such as Najee, Roger David, Country Road and Georges.

© Barbara Lynch, Angelo Pantalone and Blake Education 2009

Downloadables sold with a licence to print and photocopy

The contents of this book are covered by copyright laws in Australia and in other countries. The purchase price includes the right for you to print and photocopy these pages for your students without further payment to the publisher. For further information about how schools can use this photocopiable resource, please refer to the 'FAQ's on www.blake.com.au. Otherwise, you may only download, print, copy, store and communicate the contents of this book without permission when that is allowed by the copyright laws applicable to you. If you work in an Australian school that is covered by the education statutory licence, you may be allowed to copy, download, print, store and communicate parts of this book without permission from copyright owners. Under the education statutory licence, payments are made on behalf of schools to Copyright Agency, which distributes the payments to writers, artists and publishers. See www.copyright.com.au.

Copyright Agency
Level 12, 66 Goulburn Street
Sydney NSW 2000
Telephone: +612 9394 7600
Facsimile: +612 9394 7601
E-mail: memberservices@copyright.com.au

Printed in Australia by Ligare Book Printer



Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
www.blake.com.au

ISBN 978 1 92571 015 1



PEFC is committed to sustainable forest management through third party forest certification. For more information go to www.pefc.org

Disclaimer and Warning
The scales, distances, rates, measurements, volumes, weights and numbers used in this book are examples for mathematical instructional purposes only and should on no account be used in any real-life situations or as substitution for professional advice. Students and teachers using this book are urged to seek professional advice from appropriately qualified personnel in respect to any problems they may encounter.



Maths in the Kitchen is a collection of photocopiable worksheets which features the application of mathematical knowledge and techniques to various areas of cooking, health and food management. The worksheets are presented in the ongoing context of episodes of a TV cooking and lifestyle show hosted by celebrity chefs Ada and Lorenzo. Any information or theory required in the subject area is provided on the worksheet. Students are given an ‘Idea’ of the answer in some cases.

As chefs have risen to celebrity status, the importance of food preparation and nutrition has become more apparent. Climate change continues to play a vital role in agriculture as does food security and the reduction of waste. Obesity issues and fast foods replacing balanced meals are leading to major health concerns. It is important to manage foods in a safe and clean environment and to eat foods that are fresh and nutritious. In this fast-paced globalised world we can use ingredients from anywhere and an Australian apple could be baked in a delicious American apple pie!

The series has been designed to cover Years 9 and 10 syllabi but the worksheets could be used to motivate and add interest to any Junior or Middle School Mathematics course. Each topic is self-contained, and in most cases, an electronic calculator or spreadsheet program would be useful.

Contents

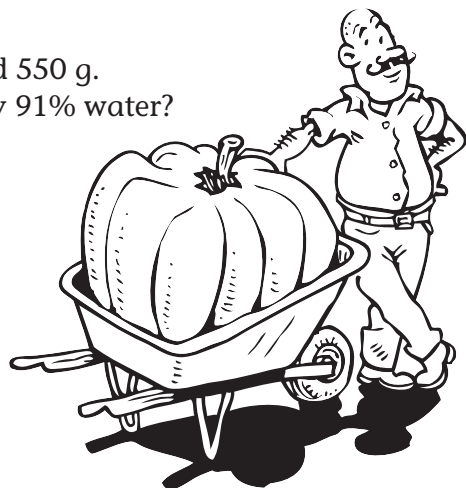
1	Funny food facts.....	4
2	Healthy and unhealthy food.....	6
3	Some delicious mathematical snacks.....	8
4	Aisle derby!.....	10
5	Fast food fantasy.....	12
6	It’s a bug’s life!.....	14
7	Wok’s wrong with that?.....	16
8	My big fat school social.....	18
9	Italian cheese cake.....	20
10	Crazy cooking capers.....	22
11	Waste or waist?.....	24
12	Paddock to plate!.....	26
13	Cooking more or less!.....	28
14	BMI – is that some sort of bike?.....	30
15	Nutrients in a lunch box.....	32
16	Global trend – overweight people.....	34
17	The nature of organics!.....	36
18	A sprouting good idea!.....	38
19	Okay, where’s the fridge?.....	40
20	It’s kool to be hot!.....	42
21	This soup is way too kool!.....	44
	Answers.....	46



Celebrity chefs Ada and Lorenzo are kicking off their new TV series with a trivia quiz night. The history of food is full of interesting and silly information and some very funny ideas about what is good to eat.

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

- 1 How old is the oldest piece of chewing gum?
Idea: 103×32 years old
- 2 What fraction of a billion dollars do young people in North America spend on chewing gum every year?
Idea: $\frac{1\ 534\ 621}{3\ 069\ 242}$
- 3 In which century was the colour of carrots changed from purple to orange?
Idea: $119 \div 7$
- 4 What fraction of the world's food supply is eaten by rats each year?
Idea: $56 \div 168$
- 5 According to research, apples float because they are made up of what percentage of air?
Idea: $\frac{1}{4}$ of $64 \times 2 - 7$
- 6 Lorenzo bought a cabbage at the market and it weighed 550 g. What mass of water did he buy if a cabbage is normally 91% water?
- 7 Ada bought a 250 g box of corn breakfast food. How much corn did she buy if corn makes up 8% of the weight of the box of breakfast food?
- 8 Mario grew a pumpkin weighing 9.08 kg. Pumpkins are generally made up of 90% water. How much pumpkin flesh in grams will Mario have to eat?
- 9 How many litres of water does it take to produce 1 kg of rice?
Idea: $7 \times 11 \times 25 + 3 \times 25 \times 4 + 25 \times 111$ L
- 10 In 1987, Giraffe Air decided to cut the cost of food on its flights to counter the higher cost of fuel. How much did it save if it used one less olive in each salad for the first class meals?
Idea: $\$ \frac{5}{7} \times 700 \times 80$
- 11 What was the mass of chocolate in a 2005 record-winning Easter egg?
Idea: $\frac{10 \times 9 \times 8 \times 7 \times 6 \times 4}{5 \times 4 \times 3 \times 2} + 2 \times 3 \times 157$ kg
- 12 When did potatoes first appear in Europe?
Idea: $\frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 4}{6 \times 5 \times 4 \times 3 \times 2} - 131 \times 2$
- 13 How many insect's legs are there in the average chocolate bar?
Idea: $\frac{2^5 \times 2^8}{2^4 \times 2^6}$



... FAST FOOD FACTS

Order of operations check!

Brackets are first,
Of is second,
Division and Multiplication,
then Addition and Subtraction,
working from left to right.



14 Coca Cola was green when it was first invented. In what year was that?

Idea: $\frac{4 \times 775 + 5 \times 1246 + 100}{5}$

15 Approximately how much lipstick does the average woman eat during her life?

Idea: $\frac{13^2 - 49}{6}$ kg

16 How many years *after* the invention of the tin can was the can opener invented?

Idea: the factors of the number are 1, 2, 3, 4, 6, 8, 12, 16, 24 and 48 and it is less than 50.

17 How many billion chopsticks are used in China each year?

Idea: 25 more than 7 more than 13

18 How many million trees are chopped down to make all the chopsticks used in China each year?

Idea: 50 less than 25 less than 100

19 How many different types of bananas are there?

Idea: more than 250 more than 200 more than 50

20 How many flowers does a bee visit during one nectar-collecting trip?

Idea: between a and b , where a is 50 less than b and b is double a .

21 How many thousands of glasses of milk does the average cow produce in a lifetime?

Idea: the number whose multiples are 2 and 2 and 5 and 10 (thousands).

22 How many litres of milk does a baby elephant drink each day?

Idea: the same as treble the number of days in September, less 10.



23 Italian villages like to compete for food records.

Work out the following records that have been claimed.

a The world's longest sandwich

Idea: $\frac{3}{4} \times 846$ m

b The world's longest torrone (nougat, honey and almond sweet)

Idea: $140.87 + 101.01 + 130.47 + 27.65$

c The world's largest diameter circular citrus tart

Idea: $21.32 - 5.37 - 6.34 - 8.41$ km

d The world's longest pizza – *Idea:* $2^4 \times \frac{3}{5} \times 25$ m

e The world's longest focaccia – *Idea:* $13.17 \times 2 + 10^2$ m

f i The world's longest sausage – *Idea:* $(9 \times 23) \times 2 - 1$ m

ii The mass of pork used in the world's longest sausage – *Idea:* $6^2 \div 4 - 1$ tonne

g i The area of the world's largest tomato-topped focaccia pizza – *Idea:* 11×3^3 m²

ii The width of the world's largest tomato-topped focaccia pizza if it was rectangular and 23.7 m long, correct to 1 decimal place.



This episode of celebrity chefs Lorenzo and Ada's TV show is all about health. They do their research around the world and find many interesting facts – and some very funny ideas – about what is good for you and what is not.

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

- 1 In what year was tomato sauce sold as a medicine?

Idea: $(7 \times 9 + 102 + 6 \times 3) \times 10$

- 2 In which century was chocolate considered to be a medicine for curing stomach ache?

Idea: $\frac{28 \times 3^2}{14}$

- 3 In which centuries was eating chocolate considered to be particularly sinful?

Idea: $\frac{68 \times 2^2}{17}$ and $\frac{272}{4^2}$



- 4 According to a survey:

- a how many million orders of French fries did Americans eat this year?

Idea: $\frac{2^7 \times 2^6 \times 2^5}{2^4 \times 2^5 \times 2^6}$

- b how many million orders of hamburgers did Americans eat this year?

Idea: $\frac{2^2 \times 3^2 \times 4^2}{2^5 \times 3}$

- 5 How many billion hot dogs do Americans eat in a year?

Idea: $\frac{2^5 \times 3^7}{2^4 \times 3^5}$

- 6 How many hectares of pizza (correct to 1 decimal place) are eaten every day in the United States?

Idea: $\pi \times 1.52^2$

- 7 a How many kilojoules does the average person burn every hour when they are:

i sleeping? – *Idea:* $2 \times 3^3 \times 7$

ii watching television? – *Idea:* $2^4 \times 5^2 - 47$

- b Hence find the difference between the number of kilojoules burnt sleeping for 8 hours and watching television for 8 hours.

- 8 a When you lick a stamp, how many kilojoules do you eat (correct to 2 decimal places)?

Idea: $\frac{0.61 \times 432}{627}$

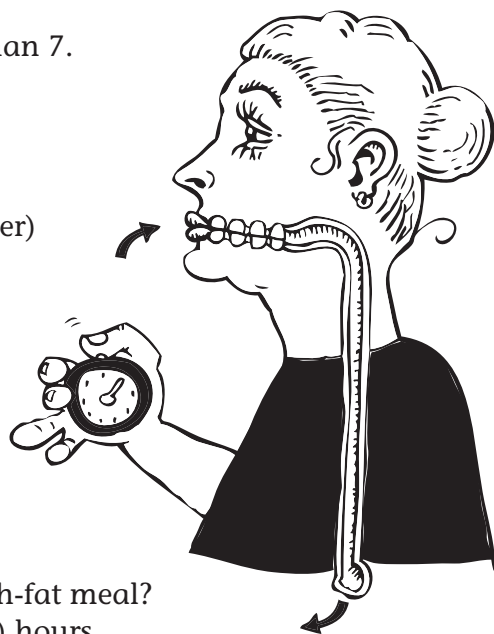
- b How many (whole number) stamps would you need to lick to eat the 880 kJ in a small serving of fries? Remember, do not round off your answer to Part a before multiplying.

- 9 Approximately how many people do not have access to clean drinking water?

Idea: $\frac{10^5 \times 10^6 \times 10^7}{10^2 \times 10^3 \times 10^4}$



- 10 How many species of capsicums are there?
Idea: 4 more than 4 more than 5 more than 6 more than 7.
- 11 Paul weighs 70 kg. How much of his body is:
 - a silver? – *Idea:* (the smallest even number) mg
 - b arsenic? – *Idea:* (5 mg more than the amount of silver)
 - c gold? – *Idea:* (one tenth the amount of silver)
 - d water? – *Idea:* (the square of the number of mg of arsenic) kg
- 12 Approximately how long does it take for food to travel down a person's oesophagus?
Idea: (between the cube of 2 and double 5) seconds
- 13 How long does it take for your stomach to digest a high-fat meal?
Idea: (the smallest number with the multiples 2 and 3) hours
- 14 How many millions of flowers do bees need to make one litre of honey?
Idea: the base of decimal numbers
- 15 How many cocoa beans are needed to make 1 kg of chocolate?
Idea: the number of minutes in 15 hours
- 16 What fraction of a teaspoon of honey can one bee produce in a lifetime?
Idea: the fraction of the clock face that the minute hand travels in 5 minutes
- 17 How many different kinds of bread are there in Germany?
Idea: more than the number of hours in 10 days, less 15
- 18 How many grapes does it take to make 1 kg of raisins?
Idea: between the number of sides on a square and the number of sides on a pentagon
- 19 How many litres of milk does it take to make 1 kg of cheese?
Idea: the same volume as that of a rectangular prism measuring 20 cm by 20 cm by 25 cm, where $1 \text{ cm}^3 = 1 \text{ mL}$
- 20
 - a What percentage of the world's people have rice as their staple food?
Idea: the fraction of the clock face the hour hand travels between midnight and 6 am
 - b How many different varieties of rice are grown?
Idea: the total length of the perimeters of ten thousand squares, each with side length 1 unit
- 21 Approximately, how many varieties of apples are there?
Idea: the number of days in 20 years of 52 weeks, correct to the nearest thousand
- 22
 - a During your lifetime, what weight of food can you expect to eat?
Idea: (the number of days in a thousand Novembers) kg
 - b How many full grown elephants is that equivalent to?
Idea: the number of sides on a hexagon

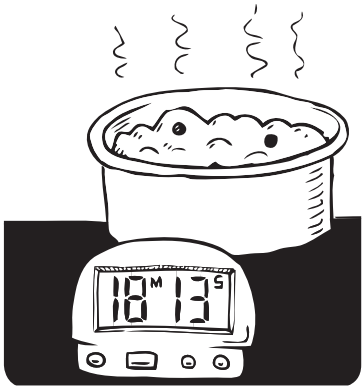




In this episode, celebrity chefs Ada and Lorenzo are cooking a number of snacks that test their mathematical ability in a range of areas – whether it be kitchen timers or scales, ingredients or recipes, cooking up a mountain or looking down a microscope. They are ready for the challenge!

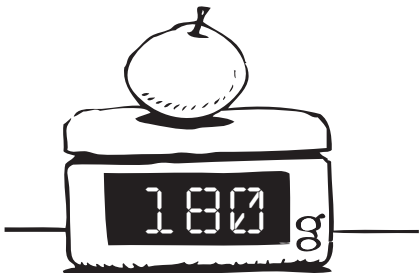
Time on your hands

- 1 A cake needs to cook for 45 minutes, and the timer shows how much time is left. For how long has the cake been cooking?

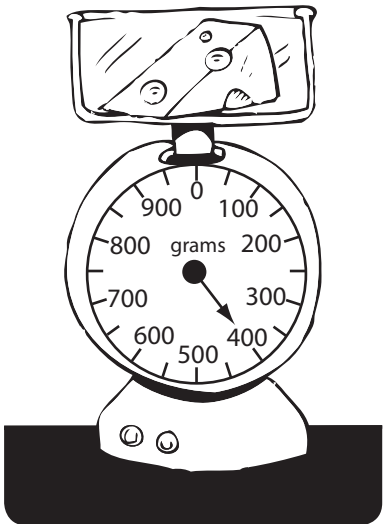


Scales

- 2 When you scale and gut a fish, you lose 5% of the weight of the fish. How much would a whole snapper weigh if the scaled and gutted weight is to be 600 g? Answer in whole number grams.
- 3 What does the apple on the digital scales weigh?



- 4 What does the cheese on the spring scales shown at the right weigh if the glass dish weighs 280 g?
- 5 Lorenzo’s family recipe for the Italian trifle, **tiramisu** (literally meaning ‘pick-me-up’), is mostly in ounces (oz). Use the fast food facts to rewrite Lorenzo’s recipe in whole number grams.



... FAST FOOD FACTS
1 gram (g) = 0.03527 ounces (oz)
1 tablespoon (tbs.) ≈ 14 grams (g)

Recipe - Tiramisu

- Ingredients (original units)**
- 4 large egg yolks
 - 4 oz marsala (sweet wine)
 - 16 oz mascarpone cheese
 - 12 oz espresso coffee
 - 2 tablespoons cocoa powder
 - 8 oz heavy cream
 - 4 oz white sugar
 - 40 lady finger cakes

- Ingredients (metric units)**
- 4 large egg yolks
 - _____ g marsala (sweet wine)
 - _____ g mascarpone cheese
 - _____ g espresso coffee
 - _____ g cocoa powder
 - _____ g heavy cream
 - _____ g white sugar
 - 40 lady finger cakes



Nice bugs!

Yoghurt and cheese are both made from fermented milk. Pickles, soy sauce and meats like salami are also fermented foods. If you put a tablespoon of live yoghurt bacteria like lactobacillus in a glass of sterile milk and keep it at 43°C, the bacteria multiply and spread through the milk to produce yoghurt in 6–12 hours.

- 6 For the bacteria to multiply, they must split in two. The table shows how one bacterium splits in two every 30 minutes. Complete the table to find how many bacteria there would be after:
- a 5 hours
 - b 10 hours.

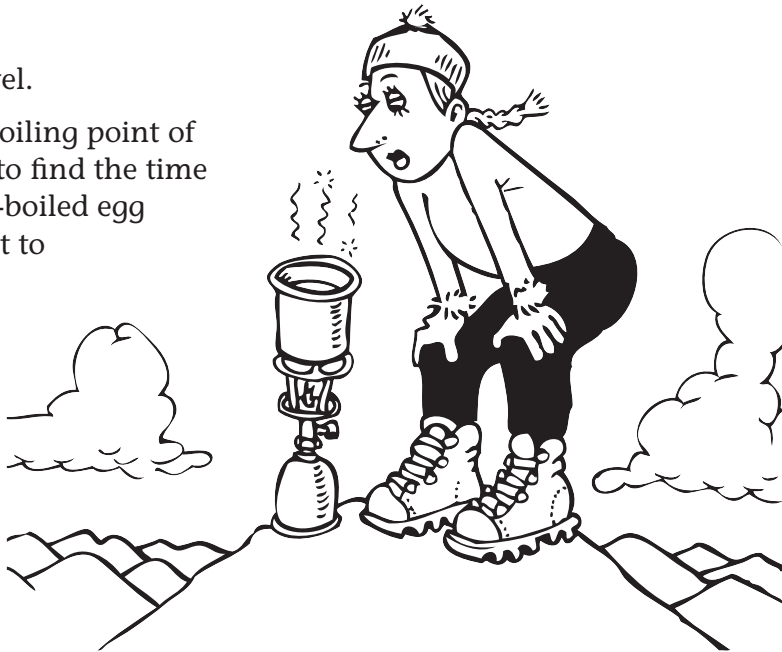
Time (mins)	Number of bacteria
0	1
30	2
60	4
90	8
120	
150	
180	
210	
240	
270	
300	

What's your attitude to altitude?

As you travel to the top of a mountain, the boiling point temperature of water reduces.

As a general rule, the boiling point temperature decreases by 1 degree Fahrenheit for every 540 feet above sea level you go.

- 7 Mount Cook is the highest mountain in New Zealand at 12 316 feet. If the boiling point is 212 degrees Fahrenheit at sea level, what is the boiling point at the top of Mt Cook in degrees Fahrenheit?
- 8 A hard-boiled egg takes 5 minutes to cook at sea level where the boiling point is 212°F. The formula:
- $E = \text{time} \times \text{temperature}$
- must always give the same value of E to cook a hard-boiled egg.
- a Find the value of E at sea level.
 - b Use this value of E and the boiling point of water at the top of Mt Cook to find the time it would take to cook a hard-boiled egg at the top of Mt Cook, correct to 1 decimal place.





In their latest TV episode, celebrity chefs Ada and Lorenzo are refereeing an all-out Aisle Derby. Our four contestants, Karina, David, Jimmy and Patricia, need to stock up their shopping trolleys in record time to win gold. They must answer the questions to work out which products to choose before moving to the checkout. Your time starts now!

To answer the following questions, choose the item that matches each description by circling A, B or C.

- 1 The largest volume of Juiciest fruit juice
 - A 1250 mL tin
 - B 1.3 L box
 - C 0.143 cL bottle
- 2 The Gluggy instant cup of soup that is cheapest per 100 g if they all cost \$2.29
 - A 42 g tomato
 - B 44 g broccoli
 - C 46 g pumpkin
- 3 The best value Tommy tomatoes
 - A 400 g for \$1.52
 - B 810 g for \$2.28
 - C 410 g for \$1.79
- 4 The lowest **total** amount of fat
 - A 300 g of ham with 14 g of fat per 100 g
 - B 250 g whole salami with 40 g of fat per 100 g
 - C 350 g of corned beef with 7 g of fat per 100 g
- 5 The lowest **total** amount of salt
 - A 300 g of ham with 1400 mg of salt per 100 g
 - B 250 g whole salami with 1570 mg of salt per 100 g
 - C 350 g of corned beef with 1380 mg of salt per 100 g
- 6 The lowest cost per 100 g of Soggy breakfast cereal
 - A 500 g packet of cornpops for \$2.39
 - B 450 g packet of cocobubbles for \$5.33
 - C 1 kg packet of wheatflakes for \$2.94
- 7 The cheapest Googy eggs per 100 g
 - A 500 g packet of one dozen eggs for \$3.26
 - B 600 g packet of one dozen eggs for \$2.82
 - C 700 g packet of one dozen eggs for \$2.50
- 8 Lightness Looker cakes with the lowest cost per point
 - A 10 point packet cherry cake for \$6.31
 - B 7 point packet carrot cake for \$6.31
 - C 8 point packet of lamington cakes for \$6.53
- 9 The Soggy breakfast cereal that is most expensive for 500 g
 - A 655 g packet of branpops for \$6.09
 - B 505 g packet of wheatflakes for \$6.09
 - C 450 g packet of cocobubbles for \$5.33





10 The cheapest Moo long-life milk per 100 mL

- A** 1 L for \$1.19
- B** 1200 mL for \$3.83
- C** 2 L for \$3.44

11 The most protein per kilojoule

- A** Pita bread with cream cheese – 903 kilojoules and 9.4 g protein
- B** Waldorf salad – 903 kilojoules and 5.2 g protein
- C** Pita bread with chicken – 1218 kilojoules and 21 g protein

EXPRESS LANE
8 ITEMS OR LESS



12 The highest fibre per gram of fat

- A** Pita bread with cream cheese – 6.5 g of fibre and 4.4 g of fat
- B** Waldorf salad – 5.5 g of fibre and 10.8 g of fat
- C** Pita bread with chicken – 6 g of fibre and 4 g of fat

13 The highest density of food in the container, where density is found by dividing the mass of the food by its volume

- A** 400 g of tomato soup in a can with volume 404 cm³
- B** 810 g of tomatoes in a can with volume 942 cm³
- C** 410 g of spaghetti in tomato sauce with volume 249 cm³

The contestants reach the checkout

14 Checkout A has a queue that is 8 people long and each person is taking 3 minutes to get through. Checkout B has a queue of 11 people, but each person is taking only 2 minutes to get through. Checkout C has only 5 people in the queue, but each person is taking 5 minutes to get through.

Which of the following is true?

- A** Checkout A will get Karina out quickest.
- B** Checkout B will get Karina out quickest.
- C** Checkout C will get Karina out quickest.

15 Checkout A has a queue that is 4 people long and each person has the equivalent of 3 shopping bags that each takes 50 seconds to check through. Checkout B has a queue of 7 people long and each person has the equivalent of 5 shopping bags, each taking 20 seconds to check through. Checkout C has a queue of 5 people and each person has the equivalent of 4 shopping bags, each taking 40 seconds to check through.

Which of the following is true?

- A** Checkout A will get Jimmy out quickest.
- B** Checkout B will get Jimmy out quickest.
- C** Checkout C will get Jimmy out quickest.



This TV episode of celebrity chefs Ada and Lorenzo's show features an exposé of fast food. Fast food is popular because it is quick, convenient and usually not expensive, but is it healthy? Let's go deep into the world of fast food to see how slippery the business can be!

1 Fat is a very efficient fuel for our bodies, providing twice as many kilojoules of energy than either carbohydrates or proteins. Of course, we need fats to keep out the cold, cushion our bodies and help build healthy cells – but we only need small amounts.

- a** A 114 g cheeseburger contains 9 grams of fat. Use the fast food facts to find the number of kilojoules of energy provided by the fat.
- b** If 30 g of fat is the total daily requirement, what percentage of the daily requirement of fat would you get from this cheeseburger?

... FAST FOOD FACTS

- 1 gram of fat gives 38 kJ of energy.
- 1 gram of carbohydrate gives 17 kJ of energy.
- 1 gram of protein gives 17 kJ of energy.
- 1 gram of alcohol gives 30 kJ of energy.

2 If Dominic wants a healthy snack, should he choose a 350 g chicken salad with 1344 kJ and 9 grams of fat or an 87 g side salad with 84 kJ and no fat with 44 mL salad dressing with 420 kJ and 6 grams of fat? Why?

Would you like fries with your salad?

3 31% of a potato chip is potato and oil, and the rest is water. If you were to buy a serve of 150 g of chips:

- a** how much would be potato and fat?
- b** how much would be water?

Give your answers in grams correct to 1 decimal place.

4 'Supersizing' means offering large serves at lower prices. A super serve of French fries at Pommegrand has 2268 kJ and 26 grams of fat. The mini serve has 60% less fat and kilojoules. How many kilojoules and how many grams of fat are there in the mini serve? Give your answers as whole numbers.

5 Chips come in various sizes. Longer chips require less oil and give more serves per kg than shorter chips. Regular chips take longer to cook but maintain their temperature for longer.

Ada decided she wanted extra-long regular chips. Use the fast food facts to find the volume of each chip if they are square in section.

Asian food might be better!

6 Some people choose Asian fast food because they believe it is healthier.

- a** Use a fast food fact to find the amount of energy in kilojoules in the 11 g of fat in a spring roll.
- b** If the fat in a spring roll provides 52% of the total energy provided by the spring roll, find the total energy in the spring roll in kilojoules.



... FAST FOOD FACTS

- Extra-long chips average 7.5 cm in length and short average 5 cm in length.
- Regular chips average 1 cm thick and shoestrings average 0.5 cm thick.



c The following table shows the fat content of four Asian dishes. Complete the table.

Food	Fat (g)	kJ in fat	Percentage of kJ as fat	Total kJ in meal
Spring roll	11		52%	
Satay chicken	76		42%	
Stir-fried chicken	59		33%	
Beef noodle stirfry	36		31%	



- 7 The recommended daily intake of salt is less than 2300 mg, of fat it is 30 g and of energy it is 8000 kJ. For the fast food meals shown in the table, find the percentage of the recommended daily intake of fat, salt and energy. Give your answers to 2 decimal places.

Meal	Fat (g)	Salt (mg)	Energy (kJ)
Chicken and chips	32	1000	3000
Hamburger, fries and milkshake	40	1400	5000
Pizza, garlic bread and can of soft drink	30	2500	4200

- 8 Your challenge is to choose the serving of French fries that has the lowest fat per gram and the highest protein per gram. Use the following survey results and a calculator or spreadsheet to make your choice.

	Mick's			ABC		Chooky		
	Small	Medium	Large	Regular	Large	Regular	Large	Family
Energy (kJ)	1050	1365	1885	1310	2860	1570	2700	3730
Fat (g)	14	18	25	13	28	16	27	37
Carbohydrate (g)	29	38	53	44	96	54	93	128
Protein (g)	2	4.2	4.5	3.5	8	4	7.5	10
Weight of serve	71	140	154	133	290	154	265	365

In 1986, the magazine *The Economist* introduced what they called 'The Big Mac Index' as a humorous guide to currency exchange rates. The idea was that exchange rates should move to keep the price of a Big Mac hamburger the same in each country. This idea has since been called 'burgernomics'.

- 9 In January 2008, a Big Mac cost \$3.45 in Australia.
- a If \$US1 = \$A1.1384 at that time, how much did the Big Mac cost in US dollars?
Answer to 4 decimal places.
- b If a Big Mac cost \$US3.57 in the United States at that time, does that mean that the Australian Big Mac cost too much or too little according to burgernomics?
- 10 A Big Mac cost 12.5 Yuan in China.
- a If \$US1 = 7.3141 Yuan at that time, how much did the Big Mac cost in US dollars?
Answer to 4 decimal places.
- b Does that mean that the Chinese Big Mac cost too much or too little according to burgernomics, if a Big Mac cost \$US3.57 at that time in the United States?



Food safety is the focus of celebrity chefs Ada and Lorenzo's latest episode. They venture into the world of the bugs that multiply in our food if we don't watch out. They know that food safety in the kitchen will stop them from getting sick.

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

- 1 a How many cases of food poisoning occur every year in Australia?

Idea: $\frac{3 \times 5 \times 7 \times 9}{189}$ million

- b If 1 in 5 cases of food poisoning is caused by incorrect food handling in the home, how many cases is this?

- 2 In the right conditions, how long does it take a single food-poisoning bacterium to grow into more than two million bacteria?

Idea: $\frac{3 \times 5 \times 7 \times 9}{135}$ hours

- 3 Food-poisoning bacteria grow and multiply fastest in the temperature range called the 'danger zone'. What range of temperatures in °C is this danger zone?

Idea: between a and b , where b is 12 times a and the difference between a and b is 55

- 4 What is the longest time you can leave high risk foods like meat, dairy, eggs, seafood, prepared foods and processed meats in the danger zone before they are no longer safe to eat?

Idea: the time it takes the minute hand of a clock to travel through 720°

- 5 At what temperature in degrees Celsius should most food be cooked?

Idea: 25 more than (25 less than double 25) plus 25

- 6 To be called 'low fat', milk must contain less than 3% fat. How much fat should there be in one litre of low fat milk?

- 7 To be called 'reduced fat', food must have a reduction in fat of at least 25% from the original product. If 'Ratrap' cheese has 48% fat, how much fat can the 'Reduced fat Ratrap' cheese contain?

- 8 To be called 'fat-free' food must have less than 0.15% fat. What percentage of this food is then 'fat free'?

- 9 A packet of potato chips was labelled as 80% fat-free. What percentage of fat is that?

A bit of this and a pinch of that!

- 10 Food additives are used to improve flavour, add colour or preserve. They can cause allergies in some people, so they are always listed on the packaging. Some additives used are listed in the table on the next page. Refer to it to answer the question below.

a Cochineal is a red colouring that comes from a beetle. What additive number is it?

b MSG is often used in Asian cooking. What additive number is it?

c How many different preservatives are listed?

d If an additive is a two hundred number, what sort of additive is it likely to be?

e How many additives are listed altogether?



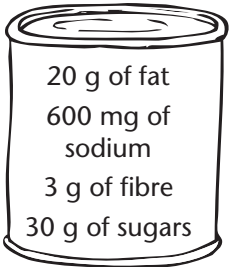


Purpose	Name	Numbers
Flavour enhancers	monosodium glutamate (MSG)	621
Food colourings	tartrazine	102
	yellow	2G107
	sunset yellow	FCF110
	cochineal	120
Preservatives	benzoates	210, 211, 212, 213
	nitrates	249, 250, 251, 252
	sulphites	220, 221, 222, 223, 224, 225, 228

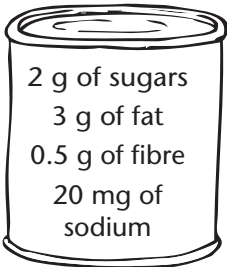


11 In Australia, all the ingredients in a food – including added water – must be listed on its label in descending order by weight. If an ingredient is less than 5% of the weight of the food then it does not need to be listed at all. Other things claimed on the label to be healthy, such as fibre, calcium or iron, must also be listed. The sugars listed include ordinary sugar as well as sugars found in other sources such as fruit. Sodium is salt.

a The ingredients on this can would be considered large amounts if they were present in 100 g of a food. Rewrite the ingredients as they should appear on a label.



b The ingredients on this can would be considered small amounts if they were present in 100 g of a food. Rewrite the ingredients as they should appear on a label.



12 Use the fast food facts to answer the following questions.

- a** On 19th February 2009, Bruno ate some cheese that had a use-by date of 12/12/08. Was he risking food poisoning?
- b** On 19th February 2009, Antonia ate some biscuits that had a use-by date of 12/12/08. Was she risking food poisoning?
- c** A loaf of bread was baked on 16th August 2008 and was labelled as best before 25/8/08. How long did Lorenzo have to eat the bread while it was at its best?
- d** Some jam was labelled with a best before date of 24/12/2008, but it should have been labelled with a use-by date of 20/11/2008. What is the difference in the time available to eat the jam?
- e** On 15th May, Kwok ate some noodles that had a use-by date of 20th April. How far out of date were the noodles?

... FAST FOOD FACTS

- If a food will last for less than two years then it must have a 'best before' or 'use-by' date.
- The use-by date is the date before which the food can be safely eaten. It is used on meat, fish, dairy and other perishable foods. Out-of-date foods cannot be sold and should not be eaten even if they look and smell OK.
- The best before date is the date before which the food keeps its quality, as long as it is stored in the right way. Out-of-date foods can be eaten safely but may not taste as good.
- Bread and meat often have a 'baked on' or 'packed on' date.



Celebrity chefs Lorenzo and Ada put together an international menu for their latest TV episode. The two-course menu comprises a Balinese entrée and an English main course. Dinner is being served at 8 pm!

... FAST FOOD FACTS



A wok is a very useful cooking vessel that appears in various cultures in Asia, Japan, India and various western countries. It is a wonderful way to stir-fry, can work as a deep fryer, a soup pot, a slow cooker and a general pan in the kitchen. It can also be used as the base for a steamer basket. The steamer shown contains two baskets in which to prepare a variety of foods.



Let's start with the entrée from Bali!

Recipe - Gado Gado

Ingredients (serves 6)

- ½ lettuce
- 250 g green beans
- ¼ cabbage
- 1 cucumber sliced
- 250 g tofu
- 2 eggs
- Peanut sauce**
- 1 red chilli
- 3 shallots
- 100 g peanuts
- 1 tablespoon brown sugar
- 1 teaspoon sesame oil
- ½ lemon
- ½ teaspoon salt

Method

Getting organised: 15 minutes

- I Fry one egg and boil the other, then slice them both.
- II Steam cabbage and green beans, and slice.
- III Steam tofu and slice. Arrange tastefully on a plate with the peanut sauce. Serving takes about 10 minutes.

Peanut sauce:

- IV Fry peanuts, shallots, chilli and sugar in 1 teaspoon of sesame oil for 3 minutes.
- V Mix and grind these with lemon and salt.

1 Steaming the beans and cabbage in the lower basket of the steamer and the tofu in the top basket takes 25 minutes. Hard boiling the egg takes 10 minutes and frying the other egg takes 2 minutes on each side. Calculate the cooking time for the recipe, including the sauce.

2 The wok holds the two steaming baskets and 500 mL of water. At a gentle boil, Lorenzo thinks that 50 mL of the water evaporates every 60 seconds.

- a How long will it take until the baskets boil dry?
- b Complete the following table, working backwards to calculate the volume of water in mL that should be used so that the steamer runs dry just as the cooking is finished, after 25 minutes.



Time (min)	0	...	18	19	20	21	22	23	24	25
Volume of water (mL)							150	100	50	0



English roast beef for the main!

Recipe - Roast beef

Ingredients (serves 6)

- 2 kg prime cut of beef
- 2 tablespoon dry mustard
- 1 teaspoon ground pepper
- 1 teaspoon pepper
- 125 g butter

Method

Getting organised: 15 minutes

- I Allow meat to reach room temperature for 2 hours.
- II Melt the butter and combine with the mustard, salt and pepper, then brush onto all the surfaces of the meat.
- III Place on a rack in a roasting pan and bake in the centre of the oven, basting occasionally until meat cooks to either rare, medium or well done.
- IV Rest for 15 minutes before carving.

... FAST FOOD FACTS

Cooking temperatures:

Rare = 170°C or 325°F

Medium = 180°C or 350°F

Well done = 190°C or 375°F

Temperature conversion:

$$^{\circ}\text{F} = \frac{5}{9} \times ^{\circ}\text{C} + 32^{\circ}$$



- 3 Lorenzo and Ada decide that this cut of meat is best cooked medium to well done (185°C).

- a At what Fahrenheit temperature will the roast be cooked?
- b The cooking time is 20 minutes per 500 g of meat. Calculate the cooking time for the roast.

- 4 Before it is cooked, the roast measured 25 cm long, 11.5 cm wide and 9 cm high. The cooking process is expected to shrink the volume of meat by 15%. What is the volume:

- a before cooking?
- b after cooking?

Answer correct to the nearest cubic centimetre, but be careful not to round off your answer to part a before calculating part b.

- 5 Dr Spec, a scientist friend of Ada's, reckons that the cooking time for meat depends on the thickness, not the weight of the roast.

'When you double the thickness of a roast it takes four times as long to cook through, and nine times as long when you treble the thickness,' Dr Spec said.

Lorenzo knows that a steak 3 cm thick takes 9 minutes to cook but their roast is 9 cm thick or three times the thickness.

- a According to Dr Spec, for how long should they cook the roast?
- b How does this time compare with the recipe book time they used in Question 3?

- 6 When it comes to cooking meat, Ada and Lorenzo have got licence to grill. They know that another factor in the cooking of meat is the temperature of the pan. An average temperature used to cook meat in a pan is 160°C.

- a When the cooking temperature was raised by 10°C, the cooking time reduced by 15%. What is the new cooking time for the steak if it took 9 minutes originally?
- b Dr Spec wants her steak medium rare. For this, the cooking time needs to be reduced by a further 9%. How long did Dr Spec's steak take to cook?

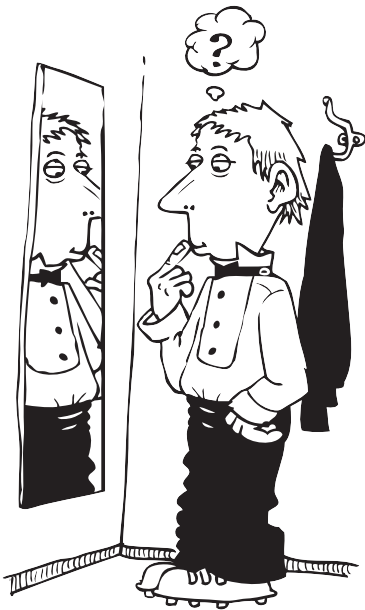


In their next episode, TV celebrity chefs Lorenzo and Ada are planning the catering for a Year 12 school social. A boy's school called St Rock is joining forces with a girl's school called St Opera to celebrate their graduation.

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

Am I on your dance card?

Lorenzo and Ada need to establish the number of students attending the social to determine the quantity of food required.



	St Rock – boys		St Opera – girls	
Year level	Number of students	% of total	Number of students	% of total
Year 7	190		210	
Year 8	210		230	
Year 9	230		225	
Year 10	220		230	
Year 11	220		245	
Year 12	225		250	
Total				

- 1 Complete the tables by calculating correct to 2 decimal places:
- a the total number of students attending St Rock.
 - b the % of each year level for St Rock.
 - c the total number of students attending St Opera.
 - d the % of each year level for St Opera.

- 2 What is the total number of students in year 12, from both schools?
- 3 Correct to 2 decimal places, what is the % proportion of:
- a boys?
 - b girls?

What's on the menu? Quick, get me the telephone!

Suppli balls are fried rice balls with a cheese centre. When mozzarella cheese is used, they are called 'suppli al telefono' (telephone rice balls) because when you bite into them, the melted cheese flows out in a long strip like the cord that dangles from the receiver when a telephone is picked up.

A recipe which serves 10 people each with 2 suppli balls includes ½ onion (minced), 500 g Arborio rice and 300 g of mozzarella cheese, cubed into 20 pieces.

- 4 Assuming the catering is for 480 people, by how much does the suppli balls recipe need to be multiplied to allow 1 serve per person?
- 5 For 480 serves:
- a how many suppli balls will be made? *Idea:* $\frac{15\ 360}{16}$
 - b how many onions will be required?
 - c how much rice will be required?



- 6**
- a** 500 grams of uncooked rice is equivalent to 2 cups. Use the fast food fact to find how many cups of cooked rice one batch of the recipe will yield.
 - b** Hence find how many cups of cooked rice there will be for the total number of suppli balls you found in question 5a.
 - c** How many grams of mozzarella cheese are there in each suppli ball? **Idea:** $\sqrt{225}$
 - d** If each cube of mozzarella cheese stretches to 15 cm when bitten into, what would the total length in metres be if the entire quantity was joined into one long rope of cheese?
Idea: the number of the school year level attending the social, squared!

... FAST FOOD FACT

Rice increases in volume by a factor of three when it is cooked.

Ahoy there! How much fish do I need to catch?

Recipe - fish and chip basket

Ingredients (serves 10)

- 1.5 kg potatoes (about 8)
- oil
- 4 kg white fish

Batter

- 2 cups all-purpose flour
- 250 mL of soda water
- 125 mL whole milk
- salt and pepper

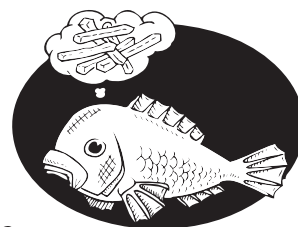
Method

- I** Scrub the potatoes and halve lengthwise. Then cut each half into 6 wedges, transferring as cut to a large bowl of ice and cold water.
- II** Heat oil to 165°C. While oil is heating, drain and dry potatoes thoroughly.
- III** Fry the potatoes until edges are just golden, about 4 minutes. Cool and then fry the potatoes again until deep golden brown and crisp, about 5 minutes.
- IV** Increase oil temperature to 190°C.

Fish batter:

- V** Sift the flour into a bowl, then whisk in soda water and milk. Pat fish dry.
- VI** Coat the fish in batter then slide into oil. Fry coated fish until deep golden, about 5 minutes.
- VII** Season the fish and chips with salt and pepper.

- 7**
- a** If 1.5 kg potatoes serves 10, what will one serve weigh in grams?
 - b** If 4 kg fish serves 10, what will one serve weigh?
 - c** Assuming there are 480 people attending the social, what mass of potatoes in kilograms will be required?
 - d** Use the ingredients list to find the approximate mass of one potato. Hence find the number of potatoes that will be needed for the social.
 - e** From the quantity and instructions for cutting the potatoes, how many chips will be obtained from all the potatoes needed for the social?
 - f** Cooking a batch of chips at 165°C takes 9 minutes. The chips are cooked in 6 batches. Use the answer to part c to find the weight of each batch.
Idea: $\frac{(66 \times 18) \times (86 - 70) \div 16.5}{24 \times 4}$ kg
- 8**
- a** For the 480 people attending the social, what weight of fish is needed?
 - b** At 190°C, the fish is deep fried for 5 minutes. Each batch being fried weighs 13.85 kg. How many batches will it take to cook all the fish needed for the social?
- 9** If multiplying the recipe by 48 will cover the quantity needed for the social, how much of each of the following ingredients is needed to make the batter?
- a** flour **b** soda water **c** milk





Gas Mark 5? How is that going to cook my cheese cake? Is it 350° Fahrenheit (F) or is it 180° Celsius (C)? In this episode, TV celebrity chefs Lorenzo and Ada are going to bake an authentic Italian cheese cake originating from Lorenzo's home town of Pescara, on the Adriatic coast.

Oven and out!

Around the world, ovens vary in the way they cook. Wood-fired ovens are complicated to use and require a lot of experience. Folklore has it that once the oven seemed hot to touch, a small fistful of flour would be thrown into the oven to see how long it took to brown, as a guide to temperature! Today's ovens are generally heated by electricity or gas, and have been built to accurately control temperature.

1 Lorenzo's Italian cheese cake is to be cooked at 350°F for 45 minutes.

a Use the fast oven facts to find the equivalent Celsius temperature in a conventional oven.

b What would be the equivalent Gas Mark setting?

c What would be the Celsius temperature in a fan-forced electric oven if it needed 45 minutes in a conventional oven?

d What would be the cooking time for the cheese cake in a fan-forced oven?



... FAST FOOD FACTS

Gas Mark is the old term for the temperature setting on an oven.

160°C = 300°F = Gas Mark 3

170°C = 325°F = Gas Mark 4

180°C = 350°F = Gas Mark 5

Fan-forced ovens circulate the hot air in the oven and so the food cooks 10% faster at a 20°C cooler temperature than a **conventional** (not fan-forced) oven.

Recipe - Lorenzo's Cheese Cake - Crostata di Ricotta

Ingredients (serves 12)

- 750 g fresh ricotta cheese
- 200 g caster sugar
- 125 g double cream
- 4 eggs (60 g each) separated
- 30 mL Sambuca liquor
- 10 g icing sugar to dust

Pastry base

- 150 g plain flour
- 60 g almond meal
- 50 g icing sugar
- 125 g cold butter, chopped
- 1 egg (60 g)

Method

Collecting all the ingredients takes about 15 minutes.

- Process the flour, almond meal, icing sugar and butter in the bowl of a food processor for 2 minutes. Add the egg and process for another 1 minute. Knead until smooth on a floured board (5 minutes). Shape into a disc and cover with a tea towel. Place in the refrigerator for 30 minutes.
- Preheat oven to 350°F for 20 minutes. Roll out the pastry to a 3 mm thick disc. Line a tin with pastry. The tin should be round with a removable base and base radius measurement of 20 cm. Be tidy.
- Cover the pastry base with baking paper and fill with pastry weights, rice or dried beans, to stop it rising. Place on a baking tray and bake in the oven for 10 minutes. Remove the paper and weights and bake for a further 10 minutes or until golden.
- Separate eggs. Beat the yolks, ½ of the caster sugar, ricotta cheese and Sambuca. Add the cream and set aside.
- Whip eggs whites until stiff, for about 4 minutes, and then add the other half of the caster sugar. Combine with the cheese mixture and place it into the tin that has the pastry and bake for 45 minutes at 350°F. Cool for 1 hour then dust with the 10 g of icing sugar. Serve.



Ready, set, cook!

- 2

Read through the five-step method and calculate how long it will take to collect the ingredients, prepare, bake and cool the cheese cake.
- 3

Answer the following questions and complete the table below with numbers correct to the nearest whole number.

a

Use the fast food facts to calculate the total number of kilojoules for each ingredient used in this recipe.

b

Hence find the total number of kilojoules for the entire recipe.

c

Using the fast food facts, calculate the total fat content for each ingredient used in this recipe.

d

Hence find the total fat content for the entire recipe, to the nearest gram.
- 4

a

The recipe serves 12. How many kilojoules per serve is that?

b

What is the fat content per serve?
- 5

Lorenzo eats his serve of cheese cake and $\frac{1}{2}$ of Ada's serve.

a

How many kilojoules did he eat?

b

How many grams of fat did he eat?
- 6

Ada finds a low fat ricotta cheese with half as much fat as the one used in the recipe.

a

Using the same quantity indicated in the recipe, what is the new total amount of fat for the cheese cake?

b

What is the new amount of fat per serve?

c

How much fat has been saved per serve by using a low fat substitute? Answer to 2 decimal places.
- 7

A chef is using the cheese cake pastry recipe to make small pastry shells for other sweet fillings. Once combined, chilled and rolled to 1 cm thick, he is able to make 15 of the 12 cm diameter circular pastry shells without wasting any.

Given that the volume of a cylinder of radius r cm and height h cm is given by $V = \pi r^2 h$, find the volume of pastry rolled out, correct to 1 decimal place, in:

a

1 pastry shell

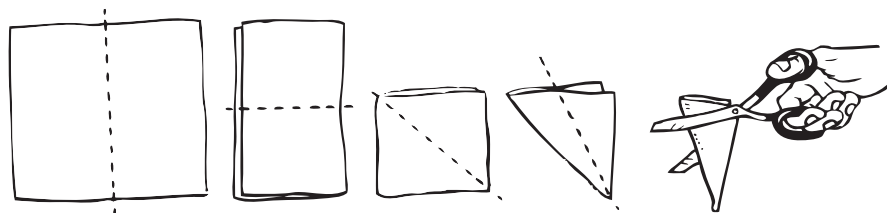
b

15 pastry shells, and hence all the pastry used.
- ... FAST FOOD FACTS
- | Food | Quantity | Kilojoules | Fat |
|--------------------------|-----------|------------|------|
| Ricotta cheese | 100 g | 565 | 10 g |
| Egg | 60 g each | 375 | 7 g |
| Double cream | 20 g | 380 | 10 g |
| Butter | 10 g | 315 | 8 g |
| Almond meal | 30 g | 730 | 16 g |
| Plain flour | 150 g | 2360 | 2 g |
| Caster sugar | 110 g | 1870 | 0 g |
| Icing sugar | 70 g | 1190 | 0 g |
| Sambuca (Italian liquor) | 30 mL | 420 | 10 g |
- | Ingredients (serves 12) | Recipe quantity | Kilojoules (kJ) | Fat (g) |
|--------------------------|-----------------|-----------------|---------|
| Filling: | | | |
| Fresh ricotta cheese | 750 g | | |
| Caster sugar | 200 g | | |
| Double cream | 125 g | | |
| Eggs | 240 g | | |
| Sambuca (Italian liquor) | 30 mL | | |
| Icing sugar | 10 g | | |
| Pastry base: | | | |
| Plain flour | 150 g | | |
| Almond meal | 60 g | | |
| Icing sugar | 50 g | | |
| Cold butter | 125 g | | |
| Egg | 60 g | | |
-
- © Barbara Lynch, Angelo Pantalone and Blake Education 2009
ISBN 978 1 92571 015 1
- Maths in the Kitchen 21



In this episode, TV celebrity chefs Lorenzo and Ada investigate the equipment used by chefs to turn out the dishes they prepare. They must get out their tape measures and find the right kitchen equipment for cooking. Mathematical capability in the kitchen is important for success.

- 1 Will a roast turkey measuring 360 mm by 280 mm by 240 mm fit in my microwave oven which has the dimensions width 31.5 cm, height 26 cm and depth 29.4 cm?
- 2 A piece of kitchen paper 20 cm square is to be used to line the bottom of a cake tin to stop the cake sticking to it. The folding is done as shown. You can try it yourself!

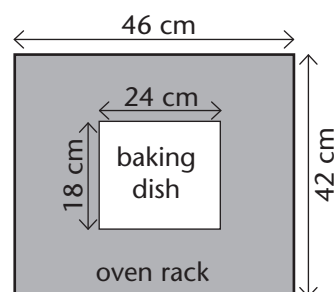


- a Approximately what shape will the paper be when you cut it as shown?
 - b Will the shape fit snugly in the bottom of a pan of radius 10 cm?
- 3 A microwave oven is advertised as having a capacity of 27 L with the interior dimensions width 359 mm, height 217 mm and depth 352 mm, and with the exterior dimensions width 510 mm, height 304 mm and depth 380 mm.
 - a Find the interior capacity (volume) of the microwave oven in litres, correct to 2 decimal places. *Hint: 1 cm³ $\equiv$ 1 mL*
 - b Is the advertised capacity correct?
 - c Find the exterior capacity of the microwave oven.
 - 4 A wall oven has a capacity of 59 L. The oven is 46 cm wide and 42 cm deep inside. How high is the inside of the oven in centimetres, correct to 2 decimal places?
 - 5 A free-standing stove is 895 mm wide and 600 mm deep. How high is it if the outside volume is 92 000 cubic centimetres?
 - 6 The turntable inside a microwave oven has a diameter of 300 mm. What is the area of the turntable in whole number mm²?

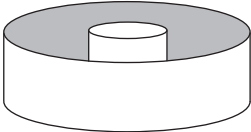
... FAST COOKING FACTS

1 litre capacity = 1000 cm³ volume
Area of a circle = πr^2 where r is radius

- 7 An electric oven measures 46 cm wide, 28 cm high and 42 cm deep inside. There are two racks that can be placed in any of the 5 equally spaced positions. The lowest position of the rack is on the bottom and the highest position is 3 cm below the top.
 - a What is the largest possible gap between the two racks?
 - b What is the smallest possible gap between the two racks?
Hint: think how many spaces there will be between the 5 positions.
- 8 a Find the length of the diagonal across the base of a rectangular baking dish measuring 24 cm by 18 cm.
Hint: use Pythagoras' Theorem.
 - b What area of an oven rack is **not** covered by the baking dish in part a if the oven rack is 46 cm by 42 cm?





- 9 Find the volume of the following cake tins.
- a rectangular tin measuring 20 cm by 8 cm by 6 cm
 - b circular tin with radius 26 cm and depth 6 cm, correct to 1 decimal place
- 10 A cylindrical cupcake tin has a radius of 3 cm and a height of 2 cm. Find the volume in cm^3 correct to 2 decimal places of 12 cupcakes, if each fills one tin. Answer correct to 1 decimal place.
- 11 A cylindrical cake pan has a diameter of 25 cm and a height of 6 cm. What is the volume of the pan? Answer correct to 2 decimal places.
- 12 A rectangular cake tin is 25 cm long, 6 cm wide and 3 cm high. The mixture is going to expand to twice its volume when it cooks.
- a How far up the side of the tin should you fill it so that it does not overflow?
 - b What will be the volume of mixture used if the cooked cake fills the tin?
- 13 1.5 L of cake mixture is placed in a cylindrical cake tin measuring 25 cm in diameter and 6 cm high. It will rise $2\frac{1}{2}$ times its depth during cooking. Will it overflow the tin?
- 14 A ring cake tin is in the form of a cylinder. It has a diameter of 25 cm and a height of 6 cm, and a centre section of diameter 5 cm that will form the hole in the cake, as shown. Find the volume of the ring cake in cm^3 correct to 2 decimal places, if it comes right to the top of the tin.
- 
- 15 Find the volume of the 0.3 cm thick icing on the top of a cupcake with 3 cm radius. Answer in cm^3 correct to 2 decimal places.
- 16 Find the area of the icing in cm^2 on a bar cake that measure 25 cm by 6 cm by 4 cm deep, if it is iced on all sides except the bottom.
- 17 A cake is made using two identical sponge cakes with a thin layer of cream between them.
- a What is the area of the cream to the nearest cm^2 if each cake is 20 cm in diameter and 6 cm high?
 - b The cake is to be iced on the top and around the side but not on the bottom. What area of icing is needed? Answer to the nearest cm^2 .

... FAST COOKING FACTS

Surface area of a cylinder
= area of circular top and bottom, plus area of side
 $= \pi r^2 (+ \pi r^2) + 2\pi rh$
where r is radius and h is height
Volume of a sphere $= \frac{4}{3} \pi r^3$ where r is the radius.

- 18 A set of 4 measuring spoons can be used to measure out $\frac{1}{4}$ teaspoon (1 mL), $\frac{1}{2}$ teaspoon (2 mL), 1 teaspoon (5 mL) and 1 tablespoon (20 mL). Each spoon is a hemisphere (half a sphere). Complete the table at the right to prove that the volume of each spoon is correct.

Radius of spoon	Volume of sphere	Volume of hemisphere (cm^3)
0.7816		
0.9847		
1.3365		
2.1216		



Celebrity chefs Ada and Lorenzo are shooting the next episode of their TV cooking show at Lorenzo's house. Picture Lorenzo's refrigerator: a carrot, some cauliflower, 6 beans, some garlic, the last of the sliced ham, one egg, half a tin of pineapple and some pretty hard cheese. Using his leftovers will save his wasted food from the rubbish tip where it would cause nasty methane gas.

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

- 1 Using leftovers for one meal allowed Lorenzo to have one 'free' meal every week. If, on average, a meal at home costs \$8.52, how much does Lorenzo save every year of 52 weeks by using his leftovers this way?
- 2 A worldwide chain of coffee houses has a policy of keeping a tap running continuously to keep their utensils clean. As a result, they waste 23 million litres of water per day. How many billions (thousand millions) of litres of water do they waste in a year of 365 days?
- 3 5 percent of a particular country's leftovers would feed the 4 million people of Costa Rica for a whole day. Botswana has a population of 1.6 million. What percentage of leftovers from this particular country would feed the people of Botswana for a day?
- 4 In 2004, the approximately 20 400 000 Australians put three million tonnes or \$5 billion worth of food in the rubbish bin. A billion is a thousand million.
 - a What mass in whole number kilograms per year is that per person?
 - b What value of food in dollars and cents per year is that per person?
 - c Correct to 1 decimal place, what mass in kilograms is that per person?
 - d Correct to 1 decimal place, what value of food in dollars and cents per week is that per person?
 - e Each week in Australia, the equivalent of one in five supermarket bags full of food goes in the rubbish and hence into landfill. If each full supermarket bag weighs 3 kg, find for each Australian, each year, the number of supermarket bags full of food:
 - i trashed
 - ii purchased.
- 5 It has been noted that nearly half of all food bought by American households is wasted.
 - a If the average American household wastes \$US 500 per year in uneaten food, how much does that household spend on food each year?
 - b If there are 115 million households in the United States, find the total amount spent on:
 - i uneaten food
 - ii food altogether.

... FAST LEFTOVER FACTS

- ☒ Don't buy too much.
- ☒ Don't 'buy one get one free' unless you need two.
- ☒ Don't shop hungry.
- ☒ Plan meals and buy only for them.
- ☒ Only buy frozen food in bulk.
- ☒ Organise your fridge.
- ☒ You can cook once and eat twice by cooking double, but remember to reheat at a high enough temperature.





- 6 On average, households in Britain throw away a third of all food they buy and half of that is edible. If these households waste between £UK250 and £UK400 of edible food each year, how much do households spend on food each year?
- 7 The amount of food wasted by the average person in Britain each year is UK£350 worth. If this amounts to 408.7 million tonnes being dumped by the 61 million people in Britain, find:
 - a what weight of food in kilograms is dumped by each person in Britain
 - b the amount of money wasted in dumped food in Britain.

Dumped food waste produces methane gas!

- 8 What weight in millions of tonnes of the world's methane emissions are produced each year by food dumped as landfill?

Idea: between a and b , where a is the even number between 29 and 31, and b is 10 times the prime number between 5 and 11. Remember, a prime number is a number whose only factors are itself and one.

... FAST FOOD WASTE FACTS

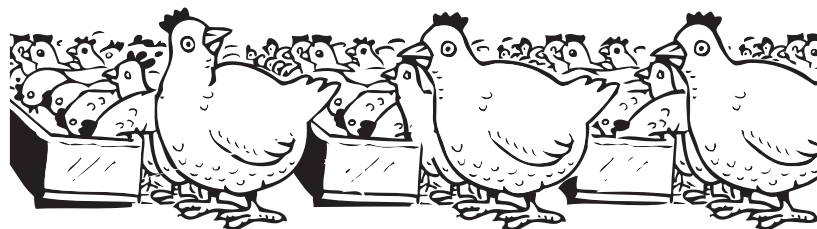
- Producing food produces a fifth of carbon emissions.
- Decomposing food releases methane.
- Most landfill methane currently comes from developed countries.
- Developed countries make more waste.

- 9 According to statistics taken in Japan, companies who manufacture and sell food, together with the restaurants that prepare food, produce 11 million tonnes of food waste each year. Also, 70% of food leftovers are recycled, half being turned into animal feed.
 - a How much food waste in whole number tonnes is recycled:
 - i altogether?
 - ii into animal feed?
 - b This means that less than 5% of the food waste is turned into the greenhouse gas, methane. How much methane in tonnes is produced?
- 10 Methane gas can be used to produce clean electricity. One million tonnes of waste dumped as landfill at Cranville produced 4 million cubic metres (m^3) of landfill gas within a year of the tip being closed. Half of the gas was methane.
 - a What volume of methane was produced by a million tonnes of waste?
 - b 1 Watt hour (Wh) is the power of 1 Watt used for 1 hour. An electric stove uses 8 thousand Watts (8 kilowatts or 8 kW) hours. How many kilowatts will the stove use if it is on for 3 hours?
 - c Power is sold by the kilowatt hour. If a megawatt hour (MWh) is a million Watt hours, how many kWh are there in a MWh?
 - d The methane gas from Cranville tip can produce 5000 MWh of electricity and this can be sold at 7 cents per kWh. How much can be earned from selling the gas from Cranville as electricity?





In this episode Lorenzo and Ada go on safari to discover how food ends up on our plates! It is important to understand the process involved in food production and handling; for instance, how seasonality and freshness contribute to nutritional value. Do you know how chicken is reared to become delicious roast chicken?



... FAST PROCESSING FACTS

- It costs chicken processors approximately \$1.40 in feed to produce 1 kilogram of chicken.
- Chicken processors on-sell chickens to retailers for approximately \$3.75 per kilogram.

Processing

The **processors** own the new-born chicks and transport them to **farmers** for rearing. Once the chickens are the right weight, the processors sell them to the **retailer**.

- 1 Use the fast processing facts to answer the following questions.
 - a The processor is rearing 100 000 organically-fed chicks to produce premium meat. When the chicken weighs 2.5 kilograms it is sold to a retailer. How much has the processor spent in feed on rearing the chickens?
 - b The processor on-sells to the retailer for \$3.75 per kilogram of chicken meat. How much income should the processor expect?

On the farm - farmers rear chickens for the processors

- 2 Use the fast farm facts to answer the following questions.
 - a Given that the usual price paid per chicken by the processor is 65 cents, how much income should the farmer expect from selling the entire flock of 100 000 chickens?
 - b If each chicken yields 1.75 kilograms of meat, how much chicken meat in kilograms has the farmer grown?
 - c If the farmer gets 37.1 cents per kilogram of chicken meat, how much will the farmer make from the flock of chickens?
 - d If the return on investment – that is, the profit – is 8.7 cents per chicken, how much has the farmer made on the entire flock?
 - e The associated cost to farm the chickens has been 91.3% of money paid by the processor. How much money is that?

... FAST FARM FACTS

- Generally a farmer is paid **65 cents** per chicken produced by the processor. A typical bird weighs about **2.5 kilograms**.
- Each chicken produces about **1.75 kilograms** of meat.
- The farmer usually gets **37.1 cents** per kilogram of chicken meat.
- Return on investment (ROI) per bird is **8.7 cents** per chicken.

In the shops!

- 3 Use the fast retail facts to answer the following question. A specialty retailer who sells organic produce purchases the entire flock for re-sale. How much retail income does 100 000 units at \$6.50 per kilogram represent?

... FAST RETAIL FACTS

Usually retailers sell whole chickens for approximately **\$6.50 per kilogram**.



It's apple-picking time - and I don't mean an MP3 player!

Most growers keep apples fresh by storing them in a controlled atmosphere (CA) room where the temperature is 32° Fahrenheit and oxygen is replaced with nitrogen to slow ripening.

- 4 What is 32° Fahrenheit in Celsius? Use the formula:

$$^{\circ}\text{C} = \left(\frac{5}{9}\right)(^{\circ}\text{F} - 32)$$
- 5
 - a An Australian orchard has 500 mature trees ready to be harvested. How many kilograms of fruit could that yield?
 - b If each apple weighs approximately 150 grams, how many apples would that be?
 - c How many kilojoules are there in the average apple?
 - d How much of the total yield of apples by weight is water?

... FAST APPLE FACTS

- The average tree will be mature and in full production after approximately 9 years.
- A mature tree bears approximately 150 kilograms of fruit.
- An apple is 85% water and 0% fat.
- An apple makes a low kilojoule, nutritious food at 220 kJ per 100 grams.
- The average apple weighs 150 grams.

How do you like them apples!

The American White House is hosting a dinner for 850 guests. The head chef is baking American apple pies using Australian apples! The apples will be air-freighted from Melbourne to Washington.

- 6 Use the fast food facts to answer the following questions.
 - a How many apples are there in each box?
 - b What is the weight of the apples in a palette?
 - c How many apples are there in each palette?
 - d 108 boxes of apples are going to the USA. Find:
 - i the number of palettes that will be needed
 - ii the weight of the apples in kilograms
 - iii the total number of apples in the consignment.
 - e The apple pie recipe indicates that 2 kg of apples are required to make a delicious pie. How many pies will our consignment of apples make?



... FAST FOOD FACTS

Air-freighting apples from Melbourne to Washington:

- Each box contains 16 kg of apples at 160 grams per apple.
- Each palette carries 27 boxes.

Who's forking the bill?

The air-freight charge includes pick-up, freight, handling, fuel surcharge, documentation fee, security, insurance, delivery to destination and GST, and is \$5.25 per kg.

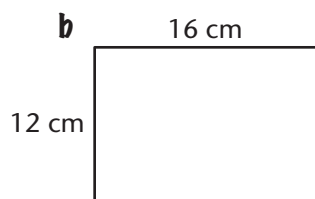
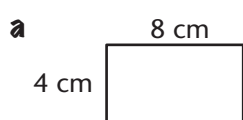
- 7 Each box of apples costs AUD\$64.00 to grow and harvest. Use your answers to Question 6 to answer the following questions.
 - a What is the cost per kg of the apples to grow and harvest?
 - b What is the cost of each apple to grow and harvest, assuming that the weight of an apple averages 150 grams?
 - c What is the total cost of growing and harvesting all the apples in the consignment?
 - d What is the total air-freight charge for the 4 palettes?
 - e What is the total cost of the apples for the White House dinner?
 - f What is the total cost of the apples per kg?
 - g What is the total cost of the apples per pie?



Celebrity chefs Ada and Lorenzo have set up a catering business, and in their latest TV episode they are being filmed hard at work, crunching the numbers. They have family recipes, and these are for family-sized batches. Sometimes they will need to scale the recipes up to cater for more serves. Sometimes they will need to scale them down to cater for fewer. Some recipes don't scale well so need to be made in smaller batches

1 Ada bought a dozen (12) buns to make hamburgers for the cricket team (eleven players). The hamburger meat makes 10 hamburgers. Can she make enough hamburgers for the team? Why?

2 Find the ratio of the areas of the tops of the cake tins shown below.



3 Lemon cordial is made by mixing lemon syrup and cold water in the ratio 1 : 4. How much syrup is needed to make up 2 L of cordial?

4 The temperature of the oven needs to be increased in the ratio 5 : 4 for a new recipe. If the temperature was 160°C, what does the new temperature need to be?

5 Recipes should not be scaled up or down more than 4 times the original serving size. A recipe for beef casserole serves 8. What are the limits of the number of servings for increasing or decreasing the recipe?

6 To make a bigger recipe in a baking pan that is twice the size, Ada needs to decrease the temperature of the oven by 15–25 degrees Fahrenheit. If the cooking temperature for the original recipe was 375 degrees Fahrenheit, at what temperature should the bigger recipe be cooked?

7 To bake half the recipe of a cake, the cooking time needs to be $\frac{2}{3}$ to $\frac{3}{4}$ of the original time. If the cooking time for the original recipe was 45 minutes, for how much time should the smaller cake be cooked?

8 I am baking half the amount of my usual apple pie. I know I will keep the temperature the same but I have to make the baking time 25–30% shorter. For how long should I cook my pie if the original recipe said to bake it for 1 hour?

9 The size of the batch affects the amount of salt needed. For double the recipe you will need $1\frac{1}{2}$ times the amount of salt. For half the recipe, you will need $\frac{1}{2}$ to $\frac{1}{3}$ times the amount of salt. The original recipe needed $\frac{1}{4}$ teaspoon of salt. Find the amount of salt needed for:

a double the recipe

b half the recipe.

... FAST FOOD FACTS

- **Batch:** the total number of serves in the original recipe
- **Scaling:** changing a recipe to give more or less than the original recipe
- **Scale factor:** the number by which you need to multiply the amount of every ingredient
- **Scale percentage:** the percentage you need to take of every ingredient



10 My recipe for a batch of pastry requires 225 g of flour, 227 g of butter, 112 g of sour cream and 5 g of salt.

- a How much does a batch of pastry weigh altogether?
- b What percentage of the total weight is:
 - i flour?
 - ii butter?
 - iii sour cream?
 - iv salt?
- c Use the fast food facts to find the amount of each ingredient if I wish to double the batch, and hence complete the table.

Ingredient	Amount in recipe	Formula percentage	Double
Flour	225		
Butter	227		
Sour cream	112		
Salt	5		
TOTAL			

... FAST FOOD FACTS

- **Formula percentage:** the weight of each ingredient, expressed as a percentage of the total weight of the batch.
- **Baker's percentage:** the weight of the key ingredient (for example, flour) is called 100% and the other ingredients are expressed as a percentage of that ingredient.
- **To scale:** scale the key ingredient and keep the relative percentages constant.

d You only have 50 g of salt. How much flour, butter and sour cream can you use for the pastry?

11 I wish to use the baker's percentage method on my pastry. (See fast food facts above.) The recipe requires 225 g of flour, 227 g of butter, 112 g of sour cream and 5 g of salt.

- a If the key ingredient is the flour and this is 100%, what are the amounts of butter, sour cream and salt as percentages of the key ingredient (baker's percentage)?
- b What do you notice about the sum of the baker's percentages?

c You are going to use 450 g of flour for your pastry.
Use the baker's percentages to find the amounts of the other ingredients, and hence complete the table.

Ingredient	Amount in recipe	Baker's percentage	Double
Flour	225	100.0	450
Butter	227	100.9	
Sour cream	112		
Salt	5		
TOTAL			

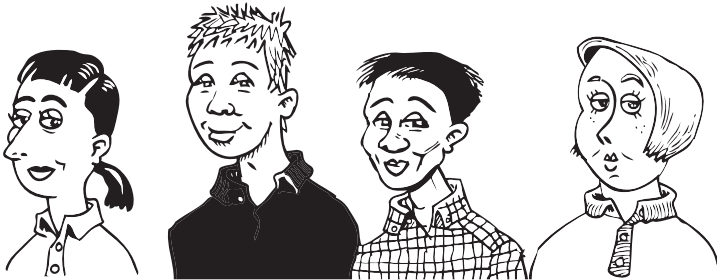
d I wish to make 1 kg of pastry. Complete the following to find the amounts of the ingredients needed.

- The original batch of pastry makes a total of _____ grams or _____% of the key ingredient.
- For a 1 kg batch I will need $\frac{1 \text{ kg}}{\text{_____ \%}} = \text{_____ g}$ of flour (100%).
- Amount of butter is _____% = _____g
- Amount of sour cream is _____% = _____g
- Amount of salt is _____% = _____g



In their latest episode, TV celebrity chefs Ada and Lorenzo are investigating ways of measuring health. The Body Mass Index (BMI) formula was developed by Belgian statistician Adolphe Quetelet (1796-1874) and was then known as the Quetelet Index. BMI is an internationally valued formula. It concerns the human body and what weight should be for height, in order to achieve good health.

Four volunteers, Karina Kalorie, David Deltoid, Jimmy Waylo and Patricia Pearsome, agree to have their BMIs calculated using the BMI formula given in the fast health facts.



... FAST HEALTH FACTS

BMI calculations require two sets of measurements: height and weight.

1 metre = 100 cm

The **metric BMI formula** takes weight measurements in kilograms and height measurements in metres.

$$\text{BMI in kg per m}^2 = \frac{\text{weight in kilograms}}{(\text{height in metres})^2}$$

Metric BMI

- 1 The first test is to find each volunteer's metric BMI.
- a Karina weighs 95 kg and is 175 cm tall.
Change her height into metres, and hence complete the following to find her metric BMI correct to the nearest whole number.

b David weighs 75 kg and is 185 cm tall. Complete the following to find his BMI correct to the nearest whole number.

BMI in kg per m² = $\frac{\text{weight in kilograms}}{(\text{height in metres})^2}$

BMI = $\frac{\text{.....}}{(\text{.....})^2}$ = kg per m²

BMI = $\frac{\text{.....}}{(\text{.....})^2}$ = kg per m²

c Hence complete the following table, correct to the nearest whole number.

Volunteer	Height (cm)	Weight (kg)	Metric BMI
Karina Kalorie	175	95	
David Deltoid	185	75	
Jimmy Waylo	175	55	
Patricia Pearsome	180	88	

Crunch time!

2 a Write each volunteer's name and BMI in the correct places in the table.

BMI standard	Weight status	Volunteer	BMI
Below 18.5	Way too thin		
18.5–24.9	Way to go!		
25.0–29.9	Clothes way too tight		
30+	Way too horizontal!		

b Who is in the healthiest range?



Waist not, want not!

- 3 Use the fast health facts to complete the following.
- a David Deltoid measures 78 cm around the waist. What is the difference in centimetres between David's waist measurement and the WCT average for men?
 - b Patricia Pearsome's waist measures 85 cm. What is the difference in centimetres between Patricia's waist measurement and the WCT average for women?
 - c Patricia's family doctor recommended a weight loss of 4 kg so she could reduce her WCT result. With a health weight loss of 0.5 kg per week, how long will it take Patricia to achieve her goal weight?

... FAST HEALTH FACTS

BMI can be misleading because fit muscular people may have high BMI because of increased muscle tone. The **Waist Circumference Test (WCT)** says that a waist measurement for men of more than 94 cm or more than 80 cm for women is unhealthy.

BEE! Like a BEE to honey!

- 4 In this exercise we will calculate the BEE of the four volunteers. Use the formulae in the fast health facts to complete the table below.

Volunteer	Age	Height (cm)	Weight (kg)	BEE	Gender
Karina Kalorie	19	175	95		Female
David Deltoid	17	185	75		Male
Jimmy Waylo	16	175	55		Male
Patricia Pearsome	18	180	88		Female

... FAST HEALTH FACTS

Basal Energy Expenditure (BEE) was created by Harris-Benedict in 1919 and is the amount of energy (in kilojoules) required to just keep your body running without your doing anything else. The difference in BEE for men and women is mainly due to the differences in body weight.

For men:

$$BEE \approx 58m + 21h - 28a + 278$$

For women:

$$BEE \approx 40m + 8h - 20a + 274$$

where m is the weight in kg, h is the height in cm and a is the age in years

Scary Spice sounds good!

Although there can be some health benefits in eating chocolate, the kilojoule and fat content can be scary – and affect your BMI!

- 5
- a A standard chocolate bar weighing 150 g contains about 3 g of protein. If 125 g of chicken yields 42 g of protein, how much chocolate provides an equivalent amount of protein?
 - b Use your answer to part a and the fast chocolate facts to work out how many grams of fat are in that amount of chocolate.
 - c Now calculate the total kilojoule intake of the amount of chocolate you found in part a.

... FAST CHOCOLATE FACTS

150 g of chocolate = 3300 kilojoules and 45 g of fat

Difficult to swallow!

- 6
- a A 300 mL can of softdrink contains approximately 4 tablespoons of refined sugar, known as 'simple sugars' or sucrose. Each tablespoon of sugar weighs 12.5 g.
 - a How many grams of sugar are there in this can of softdrink?
 - b 16 tablespoons of sugar make up one standard cup. What volume in cups does this can of softdrink contain?



In the latest episode of their TV show, celebrity chefs Ada and Lorenzo are investigating the nutrients people need to survive and grow. Nutrients are found in a large variety of foods, but they can also be taken as a supplement when this is recommended by a doctor. Lorenzo and Ada prepare lunchboxes for Patricia Pearsome, Jimmy Waylo, David Deltoid and Katrina Katorie. How nutritious are these lunch boxes?

What’s in your lunchbox? Let’s swap!

Lunchbox # 1

Patricia Pearsome’s lunchbox contains dietary fibre in 2 slices of wholegrain bread, protein in 100 g of chicken, vitamin A in 1 carrot, vitamin B2 in a 200 g tub of yogurt mixed with 1 tablespoon of wheatgerm for vitamin B6 and a bottle of tap water.

- 1 Use the nutrient table to calculate the nutrient content in Patricia’s lunchbox of:

a dietary fibre b protein c vitamin A
d vitamin B2 e vitamin B6.
- 2 Using the nutrient table below and your answers to question 1, calculate the extra nutritional intake Patricia needs to reach her RDI of:

a dietary fibre b protein c vitamin A
d vitamin B2 e vitamin B6.



... **FAST FOOD FACTS**

RDI = Recommended Dietary Intake

AI = Adequate Intake

mg = milligrams

µg = micrograms

1000 µg = 1 mg

Ada and Lorenzo’s nutrient table

Nutrient	RDI (Boys)	RDI (Girls)	Example
Dietary fibre	30 g (AI)	25 g (AI)	1 slice wholegrain bread = 3 g
Protein	64 g	46 g	125 g lean meat or fish = 42 g, 1 × 65 g egg=6.25 g
Thiamin (B1)	1.2 mg	1.1 mg	100 g pork = 0.6 mg
Riboflavin (B2)	1.3 mg	1.1 mg	200 g natural yoghurt = 0.8 mg
Niacin (B3)	16 mg	14 mg	150 g grilled fish = 12 mg
Vitamin B6	1.3 mg	1.3 mg	1 tablespoon wheatgerm = 0.35 mg
Vitamin B12	2.4 µg	2.4 µg	150 g cooked meat, beef, lamb or pork = 3 µg
Folate	400 µg	400 µg	½ cup cooked spinach = 90 µg, 1 carrot = 2000 µg

Nutrient	RDI (Boys)	RDI (Girls)	Example
Vitamin A	900 µg	700 µg	50 g cheese = 160 µg
Vitamin C	45 mg	45 mg	1 orange or 2 kiwifruits = 75 mg
Vitamin E	10 mg (AI)	7 mg (AI)	1 tablespoon sunflower seeds = 7.6 mg
Calcium	1000 mg	1000 mg	250 mL skim milk = 300 mg
Zinc	14 mg	8 mg	100 g lean meat, beef, lamb and pork = 3.6 mg
Iron	8 mg	18 mg	100 g lean meat, beef, lamb and pork = 1.1 mg
Magnesium	400 mg	310 mg	100 g prawns = 110 mg
Sodium	460–920 mg	460–920 mg	250 mL milk = 120 mg
Potassium	3800 mg	2800 mg	A 200 g potato = 900 mg



Lunchbox # 2

Jimmy Waylo's lunchbox contains 150 g of stir-fried pork, 2 kiwifruits, a 300 mL bottle of skim milk, 200 g of steamed potato with soy sauce and 1 tablespoon of sunflower seeds.

- 3 Using the nutrient table, calculate the amounts in Jimmy's lunchbox of:
 - a thiamin
 - b vitamin C
 - c calcium
 - d potassium
 - e vitamin E.
- 4 Jimmy's protein serve will also provide 3 µg of which vitamin? *Idea:* $\frac{\sqrt{576}}{2}$
- 5 How much zinc and iron does Jimmy's 150 g serve of pork provide?
- 6 What % of the maximum RDI of sodium is Jimmy getting from the milk?

Lunchbox # 3

David Deltoid's lunchbox contains 187.5 g grilled fish, 300 g baked potato, two 65 g hard-boiled eggs, 2 oranges and 75 g of cheese.

- 7
 - a Calculate David's vitamin B3 intake from the fish.
 - b Has David reached his RDI of Niacin?
 - c David doesn't eat a third of his potato. How much potassium is he getting?
 - d How much protein is David getting from the eggs?
 - e If fish contains approximately the same amount of protein as chicken, how much protein has David eaten if you include the eggs?
 - f David swapped one of his oranges for a kiwifruit. How much vitamin C did he have?
 - g David was too full to eat the cheese. How much vitamin A did he miss out on?

Lunchbox # 4

Katrina Katorie's lunchbox contains 150 g lean beef on 2 slices of wholegrain bread, 1 carrot, an omelette made with two 65 g eggs with ½ cup cooked spinach and 100 g of prawn meat, and a bottle of spring water.

- 8
 - a Assuming that there is the same amount of protein in beef as in chicken, how much protein will Katrina consume from the beef?
 - b After calculating Katrina's dietary fibre from the bread, how far away is she from reaching the RDI?
 - c How much folate is Katrina getting from the spinach?
 - d After eating the prawns, how much more magnesium does Katrina need to reach her RDI?
 - e When combining the protein values of the meat and eggs, how much over the RDI for protein is Katrina?

The RDIs are 'growing' up!

A recent study of the recommended dietary intake (RDI) of nutrients in the Australian diet increased the RDI for calcium by 25 per cent from 800 mg to 1000 mg. The review also recommended we consume twice as much folate as before, up 200 µg to 400 µg.

- 9
 - a Using the nutrient table, calculate how much skim milk contains the new calcium RDI.
 - b How many cups of spinach does Popeye need to eat to reach the new folate value?

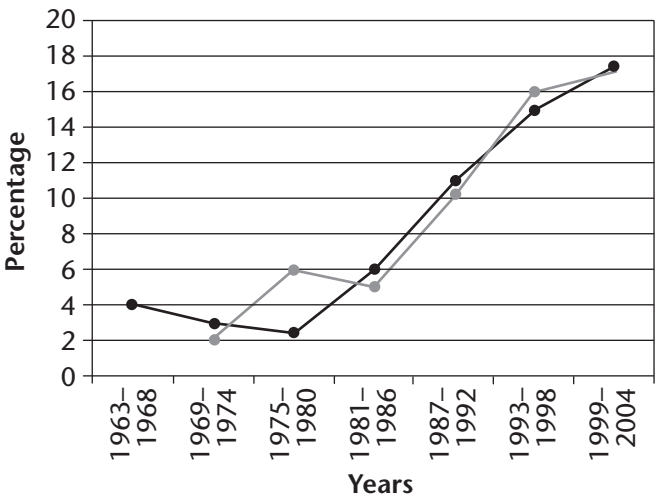


In their latest episode, TV celebrity chefs Lorenzo and Ada are preparing a report on the weights of Australian children and adolescents. First they study a large developed country called Mondo and then compare it with Australia.

Overweight and obese children and adolescents growing?

There appears to be a large number of overweight and obese children in the developed world.

1 In Mondo, Lorenzo and Ada studied the years 1963–2004, and drew the graph and table below.



— Percentage overweight 6–11 yrs (children)
— Percentage overweight 12–19 yrs (adolescents)

... FAST WEIGHT FACTS

BMI (Body Mass Index) is a number that compares weight and height.

BMI standard	Weight status	Regarded as
Below 18.5	Way too thin	Underweight
18.5–24.9	Way to go!	Healthy weight
25–29.9	Clothes way too tight	Overweight
30+	Way too horizontal!	Obese

Years	Percentage overweight 6–11 yrs (children)	Percentage overweight 12–19 yrs (adolescents)
1963–1968	4	
1969–1974	3	2
1975–1980	2.5	6
1981–1986	6	5
1987–1992	11	10
1993–1998	15	16
1999–2004	17.5	17

- a Use the table or graph to find the percentage of overweight children aged 6–11 years in 1963–1968. What was the percentage in 1999–2004?
- b For the 42 years of the survey, what was the % growth in overweight children?
- c Use the table or graph to find the percentage of overweight adolescents aged 12–19 years in 1999–2004. What was the % increase between 1987 and 1998?
- d Between 1993–1998 and 1999–2004, the change in percentage of overweight adolescents was 1%. Given this increase, what would you predict the percentage of overweight adolescents to be in 2005–2010?
- e Reading off the table, between which years did the highest % growth in the number of overweight adolescents occur?

Being overweight cost the country heaps in 2004!

- 2 Of the 250 000 000 people in Mondo, 20% were aged between 6 and 19 years.
 - a How many were there in the 6–19 years age group?
 - b If 34.5% of people in the 6–19 years age group were overweight, how many people was this?
 - c The total health cost in Mondo for 6–19 year olds in 1999–2004 was \$40 billion (thousand million). What is the average cost per adolescent?



Some BIG Australians!

... FAST FAT FACTS

When a community is overweight, it costs more to keep it healthy – medicines and hospitals cost more.

- 3 The population of children and adolescents in Australia aged 6–18 years is estimated at 6 000 000. Statistics show that 25% of children and adolescents aged 6–18 years are overweight. The current population of Australia is 21 500 000.
 - a How many overweight children and adolescents aged 6–18 are there in Australia?
 - b What percentage of the population of Australia is overweight children and adolescents?
 - c Statistics show that aerobic fitness (ability for muscles to work longer) is decreasing by 0.4% per year. Starting with 100 units in the year 2000 what would be the estimated aerobic fitness in 2010?
 - d If the overweight levels of the 6–18 years age group increased by about 35% between 1985 and 1995 with a steady increase of 3.5% per year, what would be the percentage overweight in 1990, if in 1985 the rate was 5%?
 - e If 50% of all 6–18 year olds become overweight adults, how many people would this be?

Idea: $\sqrt{562\,500\,000\,000}$

- f By 2050, the population of Australia is projected to increase to 32 500 000. If this occurs, by how much will the population increase?
- g If it is assumed that 28% of the population in 2050 will be in the 6–18 years age group, how many people will that be?
- h Assuming that the overweight percentage for the 6–18 years age group has increased to 38%, how many people will that be in 2050?



Why are we spreading out horizontally?

... FAST FAT FACTS

If you eat a lot of high energy (kilojoule) foods and don't do enough exercise to use up that energy, then you have an energy imbalance that will cause you to put on weight.

Becoming overweight can lead to major diseases like coronary heart disease, colon cancer and Type 2 diabetes.

- 4 One Australian study suggests that 20% of Australia's population is physically active, and if the physically active people increased their overall exercise by 1%, 122 lives could be saved.
 - a According to the study, how many of the current population of 21 500 000 Australians are physically active?
 - b If the physically active Australians increased their activity by 1.5% instead of 1%, how many lives could be saved?
 - c According to the study, how many Australians are inactive?
 - d If 5% of the inactive Australians increased their physical activity by 1%, how many people would that represent?
 - e The number of Australians aged 6–18 years is estimated at 6 000 000 of which 25% are overweight. If one third of overweight people increased their physical activity by 1%, how many people would that represent?

Idea: $\frac{400\,000}{10} \times \frac{1\,000}{20} \times 100\%$



In their latest episode, TV celebrity chefs Ada and Lorenzo are going green! Organic foods are foods grown according to certain standards, without unnatural pesticides, fertilisers or sewage sludge and processed without food additives. Generally they are not genetically modified (GM). Types of organic produce available include fruit and vegetables, grains, meat and meat products, honey and some processed foods.

Pick me or not?

- 1
 - a In 2008, the domestic Australian organic food market was worth \$225 million while a further \$65 million per year was received through exports. What was the combined domestic and export value of organic food in 2008?
 - b Consumer demand led to an increase in domestic sales of 12.5% in 2009. What would the Australian organic market be worth in 2009?
 - c If Australian organic food represented 1.5% of global organic food sales in 2008, what would be the global figure, rounded to the nearest billion?
- 2 From 2008, the growth in the total value of organic food produced in Australia is expected to be about 30% annually. Use your answer to question 1a to find the total dollar value you would expect it to be worth in:
 - a 2009? b 2010?
 - c 2011? d 2012?

... FAST FOOD FACTS

- From 1990 to 2008, organic food production in Australia grew at about 20% per year.
- Organic farming can protect the environment and ecosystems.
- Rotating crops keeps soil nutrients.
- 1 acre = 0.4047 hectare (hA)
- Organic fruits and vegetables cost 15–60% more than non-organic, on average.

Add one part earth, one part seedling and one part water, then wait!

- 3 Old George, the Australian farmer, uses traditional farming methods to grow organic fruits and vegetables on his 150 acre farm.
 - a How many hectares in area is George's farm, correct to 1 decimal place?
 - b Ten acres of George's land is devoted to organic herbs. What area in hectares is this?
- 4
 - a George's neighbour, Gwen, uses 4 hectares of her farm to rear jersey cows. Gwen needs 10 acres to feed 50 cows. How many cows can Gwen feed?
 - b 50% of Gwen's cows produce milk. How many cows is that?
 - c Each of Gwen's milk-producing cows is milked twice a day and yields 1.5 litres of milk per session. The cows are milked 52 weeks of the year. How much milk do Gwen's cows produce each year?
 - d Non-organic milk is sold at \$0.40 a litre, but Gwen's milk is sold at 15% more per litre. How much money will Gwen make from selling milk each year?



Climate change and organic food production

- 5 There is a drought in Far North Queensland which has affected the price of bananas. They were \$5.99 per kg before the drought, but the price has increased by 37%. What is the new price?



- 6
 - a Australian ginger is available from the grower for \$16.50 per kg. If Securit Supermarkets needs 250 kg, how much will they pay for the order?
 - b A shipment of Vietnamese ginger can be imported to Australia for \$12.50 per kg. What will Securit Supermarkets pay for 250 kg of imported ginger?
 - c What is the percentage difference in the price paid for 250 kg of Australian ginger and 250 kg of Vietnamese ginger?

Lorenzo's oil but Ada's know-how!

As far back as 3000 BC, olives were grown commercially in Crete. Today, the Greeks, Italians and Spanish make up the world's largest olive oil makers but Australia is going for green when it comes to organic olive oil.

- 7 Ada can grow 80 000 trees on a 240 hectare plantation. A standard olive tree yields 3000 olives a year, with each olive weighing about 3 g.
1 acre = 0.4047 hectare (Ha)



- a How many acres is Ada's plantation?
Answer correct to 2 decimal places.
 - b 80 000 olive trees will yield approximately how many olives in one year?
 - c What is the total weight of olives in kg yielded by 80 000 olive trees?
 - d It takes about 5 kg of olives to make 1 litre of olive oil. Using your answer to part c, how many litres of oil can Lorenzo and Ada expect?
 - e The market value for Lorenzo and Ada's oil is A\$15.00 a litre. Using your answer to part d, what is the market value of the oil?

Tracing minerals?

- 8 A study indicates that the levels of trace minerals in fruits and vegetables fell by up to 76% between the beginning of 1940 and the beginning of 1990. There is growing evidence that organic fruits and vegetables generally contain more nutrients than non-organic food. A soil association conducted a review of the evidence comparing the vitamin and mineral content of organic and conventionally grown food. It was found that, on average, organic food contains higher levels of vitamin C and other essential minerals.
 - a In the study of trace elements, how many years are there between the two dates?
 - b The study indicates that the amounts of trace elements fell by 76% during that time. Assuming that trace elements dropped by an equal percentage each year, what is that percentage?
 - c 100 grams of non-organic orange flesh contains about 54 mg of vitamin C. How much vitamin C would 225 g of this orange flesh contain? Answer correct to 1 decimal place.
 - d If 225 g of organic orange flesh contains 45% more vitamin C than 225 g of the non-organic orange flesh, how much vitamin C is that? Answer correct to 1 decimal place.
 - e A green non-organic kiwifruit contains 98 mg of vitamin C per 100 grams. If the organic kiwifruit contained 60% more vitamin C, how much is that per 100 grams?

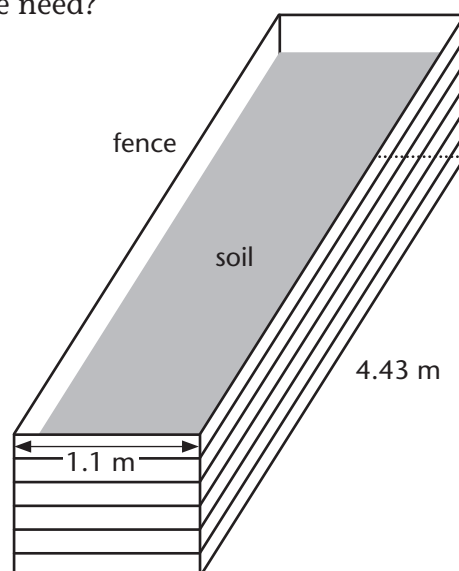
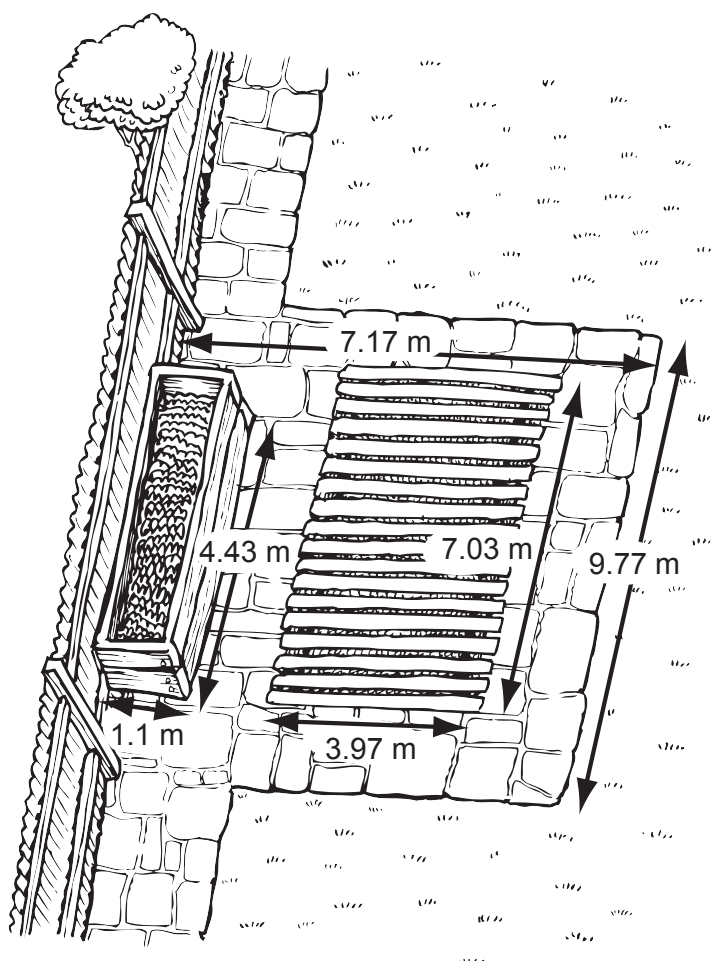


In this episode, TV celebrity chefs Ada and Lorenzo will show their viewers how to design and build a kitchen garden, so that they have a good supply of fresh herbs and vegetables for their cooking.

They have decided to build a garden in a box on their patio.

Designing and building the garden

- 1 Ada and Lorenzo's tiled patio is rectangular, as shown. Find the area correct to the nearest m^2 of the patio, excluding the paths leading in and out of it.
- 2 Timber decking is placed within the patio to allow rain water to reach the ground below. Use the measurements shown to find the area of the timber decking, correct to the nearest m^2 .
- 3 The box that will house the kitchen garden is 4.43 m long, 1.1 m wide and 60 cm high. It will be filled with composted soil to a depth of 50 cm.
 - a What is the volume of the whole garden box in m^3 , correct to 1 decimal place?
 - b What will be the volume of soil in the garden box in m^3 , correct to 1 decimal place?
- 4 The gardening book told Lorenzo that he should use 2 parts compost to every 1 part sand and 1 part of the mineral vermiculite.
 - a If Lorenzo was making 4 cubic metres of garden soil, how much compost, sand and vermiculite would he need?
 - b If Lorenzo was making 8 cubic metres of garden soil, how much compost, sand and vermiculite would he need?
 - c For the volume of soil in the box found in question 3b, how much compost, sand and vermiculite will Lorenzo need?
 - d Lorenzo will also need 1% wood ash and 0.25% lime. How much soil altogether will he have made?
- 5 The box is constructed of planks 10 cm wide. It is built against the existing fence. What length of planking has been used to make the box? Answer to the nearest metre.
Hint: Notice that it is built against the fence!





When will we plant?

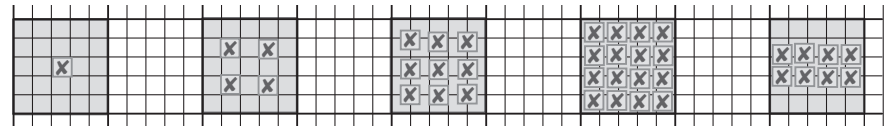
6 Different plants must be planted at different times of the year, and grow for different amounts of time before they can be picked. Ada has collected the information in the table so that she can work out when the vegetables will be ready for picking. Complete the table below for her.

Plant	Sowing date	Time before picking (days)	Date to be picked
tomato	1st November	135	
radish	1st October	30	
capsicum	1st November	135	
broccoli	25th October	135	

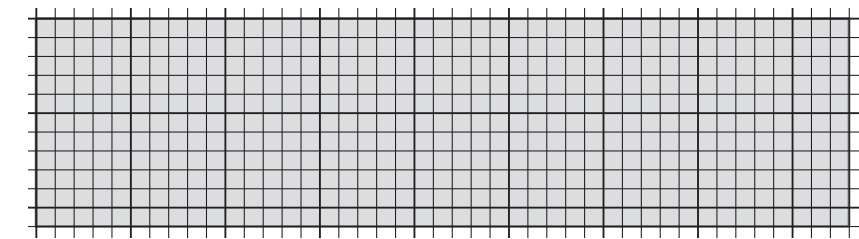


What to grow?

Ada and Lorenzo are ready to plant their vegetables. Different vegetables take up different amounts of space. Ada has collected the information about the number of plants that will fit in a square measuring 2500 cm², as you can see in the table at the right. She then drew the 2500 cm² squares on graph paper to represent the spacing of each type of plant, as you can see below.



Ada also drew the garden bed on graph paper to the same scale, as you can see below.



Plant	Number per 2500 cm ²
cabbages	1
broccoli	1
peas	1
cauliflower	1
lettuces	4
tomato	1
capsicum	1
silverbeet	4
chinese broccoli	4
celery	8
beetroot	9
garlic	9
carrots	16
spring onions	16

- 7 Using the table and Ada’s drawings on graph paper, how many plants would Ada need to fill her garden bed with:
- a cabbages? b lettuces? c garlic? d carrots?
- 8 If Ada and Lorenzo decided to grow peas along the fence, how many pea plants could they grow?
- 9 What is the maximum number of plants they could grow in the garden bed if they must grow celery, garlic, at least 12 lettuce plants and at least 10 tomato plants and cannot grow any other types of plants?

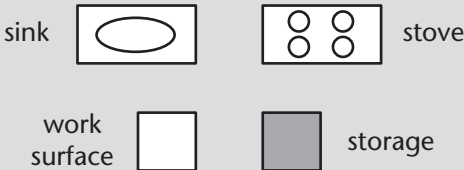


In their latest episode, TV celebrity chefs Lorenzo and Ada present the ins and outs of kitchen design, looking at different kitchen plans to see which will suit them best. They want their kitchen to have plenty of storage and work surfaces and they don't want to walk too far while they are preparing their meals.

1 Lorenzo and Ada have a choice of four different kitchens: the galley (like in a ship), the L-shaped, the U-shaped and the island kitchen with a free-standing bench. Use the fast kitchen planning facts to write the unlabelled dimensions onto each plan.

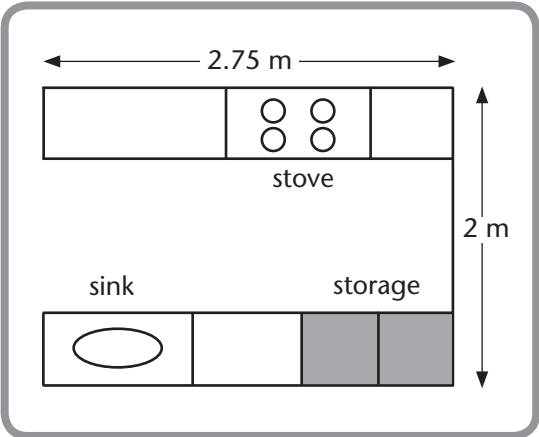
... FAST KITCHEN PLANNING FACTS

Activity zones include washing, cooking, working and storage. In the plans, these are shown using different icons:

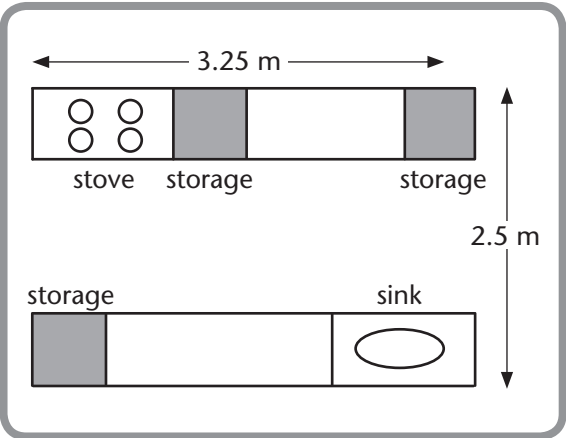


For simplicity, the benches are all 0.5 m deep, and the sink and stove are 1 m wide and 0.5 m deep.

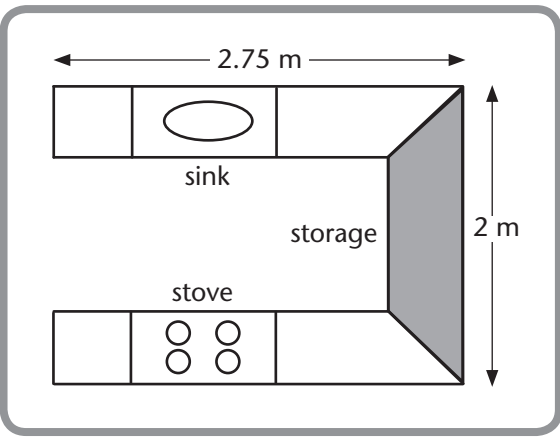
a Galley kitchen



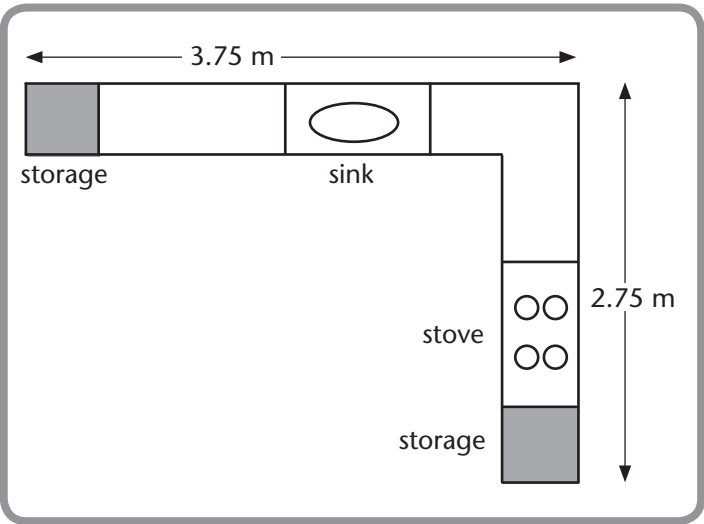
c Island kitchen



d U-shaped kitchen



b L-shaped kitchen

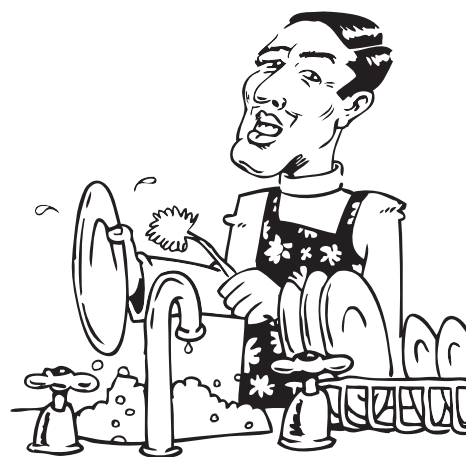


2 Use your answers to Question 1 to find the following for each kitchen. You could draw up a table to record your answers.

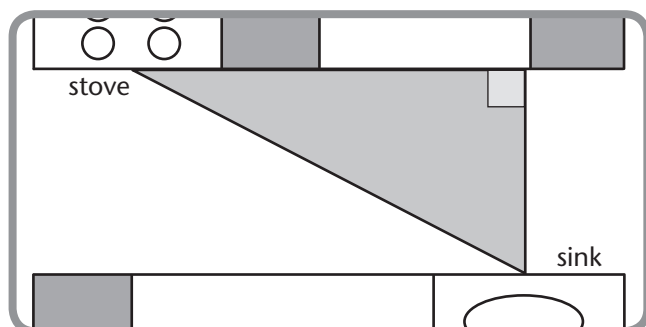
- a the total outside area
- b the area taken up by the stove
- c the area taken up by the sink
- d the area available for storage
- e the area of work surfaces
- f the volume of storage, if the height of each storage cupboard is 1.8 m
- g the area of tiling needed for the kitchen floor



- 3 a Which kitchens have the most storage?
b Which kitchens have the least storage?
- 4 When Lorenzo is cooking, he has to walk between the stove and the sink many times. Given that the kitchen plans are drawn to scale, *estimate* on which plan the distance between the stove and the sink is:
 - a the shortest.
 - b the longest.
- 5 How far does Lorenzo need to walk from the stove to the sink in the U-shaped kitchen?

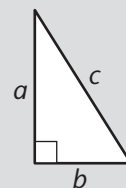


- 6 Lorenzo rather likes the island kitchen and wants to work out the distance between the stove and the sink. He draws a right-angled triangle and shades it black as shown. Find the dimensions of this triangle, and hence use Pythagoras' Theorem to find the distance between the stove and the sink correct to 3 decimal places.



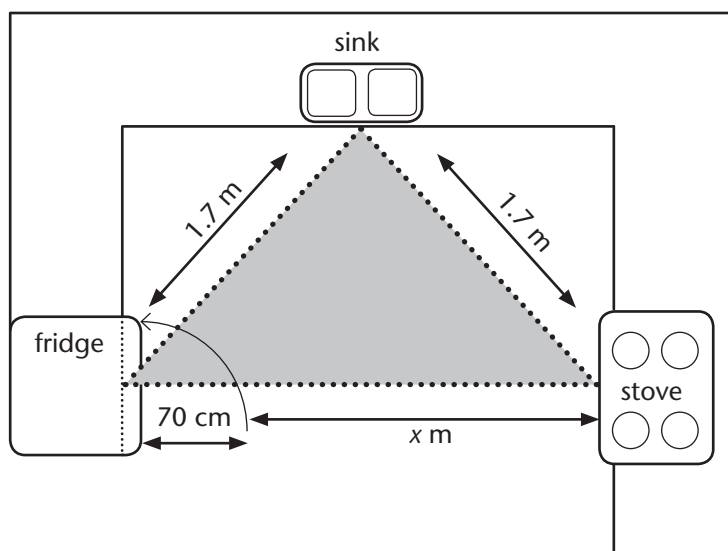
... FAST KITCHEN FACTS

Pythagoras' Theorem for a right-angled triangle is $c^2 = a^2 + b^2$.



In the 1950s, the work triangle was used to design new kitchens. The idea was that the distance between the sink, stove and refrigerator should be small enough that it did not take too many steps to get from one to another, but not so small that there was no room to work.

- 7 In the plan shown at the right, the triangle is right-angled at the sink.
 - a Use Pythagoras' Theorem to find the distance from the stove to the refrigerator.
 - b If the refrigerator door is 70 cm wide, find the distance from the opened refrigerator door to the stove, marked on the plan as x m.
 - c Use Pythagoras' Theorem to find the perpendicular height of the triangle from the sink to the unknown side.





In their latest episode, TV celebrity chefs Ada and Lorenzo are helping out an Australian high school's Year 12 students, who will set up a mobile soup kitchen in the central business district of their city to raise funds for a charity. Ada and Lorenzo have been brought in for their Maths and cooking skills!

The soup Ada and Lorenzo have chosen is the Italian vegetable soup called *minestrone* which in Italian means 'to serve up' or 'wait on'. Lorenzo will work on the recipe and Ada will work on the calculations.



Minestrone

Ingredients (serves 10)

- 500 g peeled tomatoes
- 1 kg potatoes
- 250 g pumpkin
- 2 onions
- 2 carrots
- 2 celery sticks
- 500 g cooked kidney beans
- 1 zucchini
- 2 cloves garlic
- 1 litre chicken stock
- ¼ cup olive oil
- ½ cup Italian parsley

- 1 a When cooked, this recipe makes approximately 2.5 litres of soup. If 250 mL is approximately 1 cup, how many cups of soup will they get from one batch?
- b They need 600 serves. How many litres of soup is this?
- c How many cups of soup will the 600 serves make?
- d Calculate the quantity of each ingredient required for 600 serves. Fill in your answers in the first column of the table at the right.
- e Calculate the cost for each ingredient required for 600 serves. Fill in your answers in the last column of the table.
- f What is the total cost for the 600 serves of soup?
- g The soup kitchen aims to sell one 250 mL serve for \$1.50. How much revenue will they receive if they sell all the soup?
- h After subtracting the cost of making the soup, how much money will go to the charity?
- i Australia has approximately 4500 high schools and each one participated in a *How Kool is my Soup Kitchen minestrone extravaganza*. How much money could be raised?
- j 5% of the schools raised one fifth of the total money. How much did those schools raise altogether?

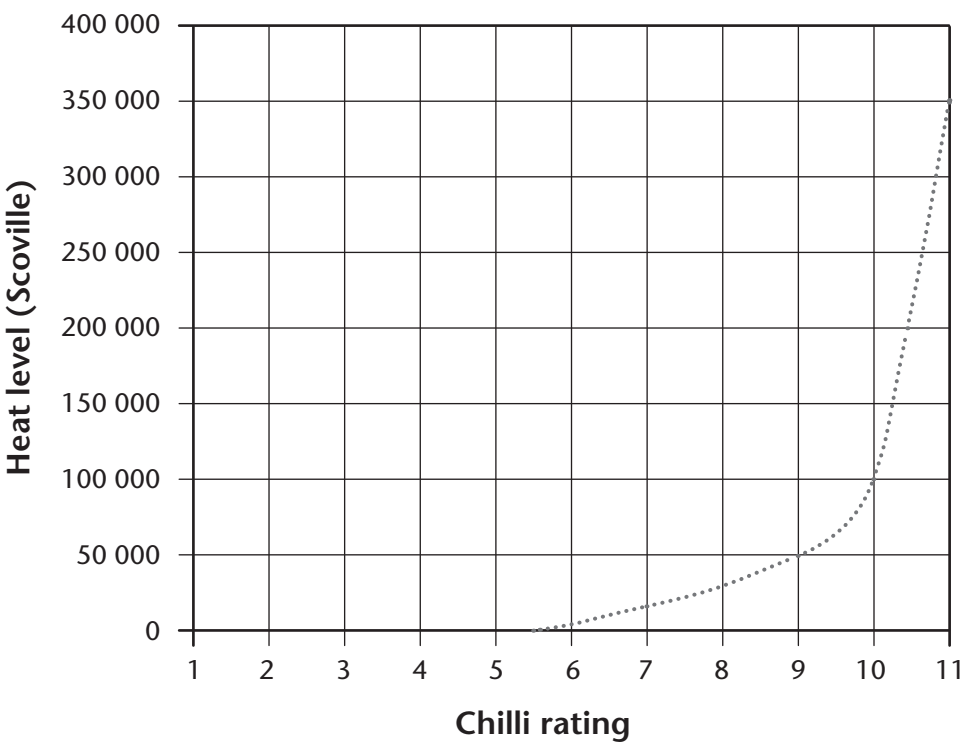
Serves 10	Quantity for 600 serves	Cost for 10 serves	Cost for 600 serves
500 g peeled tomatoes		\$0.60	
1 kilo of potatoes		\$1.20	
250 g pumpkin		\$0.50	
2 onions		\$0.20	
2 carrots		\$0.20	
2 celery sticks		\$0.20	
500 g cooked kidney beans		\$1.50	
1 zucchini		\$0.75	
2 cloves of garlic		\$0.50	
1 litre of chicken stock		\$2.00	
¼ cup olive oil		\$0.75	
½ cup Italian parsley		\$0.25	
Total cost			



Chilli Pepper Contest

The heat in chillies is often measured in Scoville units. These units are named after Wilbur Scoville who, in 1912, used tasters to tell him how much sugar and water was needed to get rid of the heat of a particular chilli. Today, scientific methods are used to measure the capsaicin which causes the heat in chillies.

- 2 In the Chilli Pepper Contest there are three categories of award – *Wimpy*, *Scorching* and *Screaming* – as well as the top award of the *Chilli Cup*.
- a Wimpy recipes have a chilli rating of 7, Scorching have a rating of 9 and Screaming have a rating of 10. Use the graph to find the number of Scoville units for each of these three categories.



- b A high profile panel of expert judges will decide the 1st, 2nd and 3rd places in each category. How many different prizes will be awarded altogether?
- c Winners in each category receive a \$6 000 voucher for kitchen appliances with second and third prizes being a half and a third of that respectively. The Chilli Cup comes with a \$10 000 cash prize. What is the total value of the prizes?
- d There are 120 cooks competing in the two-hour, action-packed finale of the competition. What are the chances of one of these cooks winning:
- i the Chilli Cup?
 - ii a prize?





On their previous episode, TV celebrity chefs Ada and Lorenzo featured a Chilli Pepper Contest and a Soup Fundraiser. These events proved extremely popular, and so a new TV show, 'How Kool is my Soup Kitchen', is on the cards for England, Indonesia and Australia. The TV network wants to forecast its popularity in the first season, and will use the TARP system to do this.

TARP? Are we talking tents?

The target audience age group for the new TV show is 26 to 44 year olds. The TV network wants to find the percentage of that target group who will watch the show at any one time. The ratings company will collect the information on what is being watched from the televisions of a sample of viewers. They will then find out how many people are watching.

Bali

- 1 a The population of Bali is 2.5 million and the target audience of 26–44 year olds comprises 42% of the population. What is the size of the possible audience for this group?
- b Research shows that the Balinese TARP for the show is 35%. That means that 35% of the target audience is likely to watch the show. How many people in Bali are likely to watch the show?
- c What percentage of the total Balinese population is that?

... FAST TV SHOW FACTS

The **reach** of a show is the number of viewers of that show each week.
TARP stands for **Target Audience Rating Point**. One TARP represents 1% of the target audience who have watched a show on TV once.

Could the Queen of England be watching?

- 2 a London has a population of 7 512 400. Complete the table by calculating the number in each age group.
- b How many people are there in the target group in London?
- c What percentage of London's population does this figure represent?
- d The English TARP for the show is 42% of the target group. How many people in London will watch the show? (What is the **reach**?)
- e What % of London's population does this represent?
- f Surprisingly, 30% of the 66+ age group tunes into the show.
 - i How many people is that?
 - ii Research has shown that in this age group, $\frac{2}{3}$ of the viewers are female. How many males in this age group are watching the show?
- g Use your answers to parts d and f to find the total number of viewers in London.
- h What % of London's population does this represent?
- i 20% of females in the 66+ age group are watching the show. Does this mean that there is a 1 in 5 chance that the Queen is watching the show? Why? (She was born in 1928.)

Age group (years)	%	Population
<5	4	
5–15	9	
16–25	16	
26–34	18	
35–44	23	
45–65	16	
66+	14	
Total	100	7 512 400





Australia

- 3 a Complete the table by calculating the percentage of the Australian population in each state.
- b If the target age group of 26–44 year olds represents 40% of the population, how many people is that?
- c The Australian TARP for the show is 45% of the target group. How many people in Australia will watch the show? (What is the reach?)
- d What percentage of the Australian population is this?
- e Given Australia’s multicultural position, the show reach has been most successful in New South Wales with 4% and Victoria with 5%. Calculate the reach in:
- i New South Wales. ii Victoria.

Australian population

State	'000 people	% of population
New South Wales	6927.0	
Victoria	5246.1	
Queensland	4228.3	
South Australia	1591.9	
Western Australia	2130.8	
Tasmania	495.8	
Northern Territory	217.6	
Australian Capital Territory	343.1	
Total	21 180.6	100.00

How kool is this TV show? A second series, stay tuned!

The first series has been very successful with advertisers keen to spend their money on the next series. The TV network wants to work out how many more viewers they are likely to attract with an increase in advertising dollars. The table shows what is currently spent on advertising.

Spending on advertising

Place being aired	Current amount spent	Current audience reach as % of population	Forecasted amount to be spent
Bali (Indonesia)	R1 000 000 Rupiah	14.70%	R1 500 000 Rupiah
London (England)	£250 000 UK pounds	17.22%	£375 000 UK pounds
Australia	\$1 500 000 Australian dollars	18.00%	\$2 000 000 Australian dollars

- 4 Using the information in the table, calculate the percentage increase in the amount to be spent by advertisers in:
- a Bali b London c Australia.
- 5 The network hopes that, with the increased advertising, the number of viewers will increase by 4% in Bali, 3% in London and 2.5% in Australia. Find the total audience expected in:
- a Bali, if the audience for the first series was 367 500
- b London, if the audience for the first series was 1 293 635
- c Australia, if the audience for the first series was 3 812 508.
- 6 If the number of viewers increases in the same proportion as the amount spent on advertising increased, what would the audience for the new series be in:
- a Bali? b London? c Australia?



1 Funny food facts

- 1 9000 years old 2 $\frac{1}{2}$ 3 17th 4 $\frac{1}{3}$ 5 25 6 500.5 g
7 20 g 8 908 g 9 5000 L 10 \$40 000 11 1950 kg
12 1586 13 8 14 1886 15 20 kg 16 48 17 45
18 25 19 more than 500 20 between 50 and 100
21 200 (thousands) 22 80
23 a 634.5 m b 400 m c 1.2 km d 240 m e 126.34 m
f i 413 m ii 8 tonne g i 297 m² ii 12.5 m

2 Healthy and unhealthy food

- 1 1830 2 18th 3 16th and 17th
4 a 8 b 6
5 18 6 7.3
7 a i 378 ii 353 b 200 kJ
8 a 0.42 b 2094
9 1 thousand million 10 26
11 a 2 mg b 7 mg c 0.2 mg d 49 kg
12 8–10 seconds 13 6 hours 14 10 15 900 16 $\frac{1}{12}$
17 more than 225 18 4 to 5 19 10
20 a 50% b 40 000
21 7000
22 a 30 000 kg b 6

3 Some delicious mathematical snacks

- 1 26 minutes 47 seconds 2 632 g 3 180 g 4 120 g
5 113 g marsala; 454 g mascarpone cheese;
340 g espresso coffee; 28 g cocoa powder;
227 g heavy cream; 113 g white sugar
6 Table: 16; 32; 64; 128; 256; 512; 1024
a 1024 b 1 048 576
7 189°F
8 a 1060 b 5.6 minutes

4 Aisle derby!

- 1 B 2 C 3 B 4 C 5 B 6 C 7 C 8 A 9 B 10 A
11 C 12 C 13 C 14 B 15 A

5 Fast food fantasy

- 1 a 342 kJ b 30%
2 Salad has lower kJ and lower fat, even with the
salad dressing.
3 a 46.5 g b 103.5 g 4 907 kJ; 10 g 5 7.5 cm³
6 a 418 kJ b 804 kJ c See table below.

Food	kJ in fat	Total kJ in meal
Spring roll	418	804
Satay chicken	2888	6876
Stir-fried chicken	2242	6794
Beef noodle stirfry	1368	4413

7

Meal	Fat %	Salt %	Energy %
Chicken and chips	106.67	43.48	37.50
Hamburger, fries and milkshake	133.33	60.87	62.50
Pizza, garlic bread and can of soft drink	100	108.70	52.50

8 Medium serve from Mick's

9 a \$US3.0306 b too little

10 a \$US1.7090 b too little

6 It's a bug's life!

- 1 a 5 million b 1 million
2 7 hours 3 Between 5°C and 60°C 4 2 hours

- 5 75°C 6 Less than 30 mL 7 36% 8 99.85% 9 20%
10 a 120 b 621 c 15 d Preservative e 20
11 a sugars, fat, fibre, sodium b fat, sugars, fibre,
sodium
12 a Yes b Yes c 9 days d 34 days e 25 days

7 Wok's wrong with that?

- 1 42 minutes
2 a 10 minutes b See table below.

Time (min)	0	...	18	19	20	21	22	23	24	25
Volume of water (mL)	1250	...	350	300	250	200	150	100	50	0

- 3 a 365° F b 1 hour and 20 minutes
4 a 2588 cm³ b 2199 cm³
5 a 81 minutes or 1 hour and 21 minutes b much
the same
6 a 7.65 minutes b 6.96 minutes

8 My big fat school social

1 a-d

	St Rock – boys		St Opera – girls	
Year level	No. students	% of total	No. students	% of total
Year 7	190	14.67	210	15.11
Year 8	210	16.22	230	16.55
Year 9	230	17.76	225	16.19
Year 10	220	16.99	230	16.55
Year 11	220	16.99	245	17.63
Year 12	225	17.37	250	17.99
Total	1295	100.00	1390	100.00

- 2 475
3 a 47.37% b 52.63%
4 48 times
5 a 960 pieces b 240 c 24 kilos
6 a 6 cups b 288 cups c 15 g d 144 metres
7 a 150 g b 400 g c 72 kg d 187.5 g; 384
e 4608 chips f 12 kg
8 a 192 kg b 14 batches
9 a 96 cups b 12 litres c 6 litres

9 Italian Cheese Cake

- 1 a 180°C b Gas Mark 5 c 160°C d 40.5 minutes
2 3 hours 22 minutes
3

Ingredients (serves 12)	Recipe quantity	Kilojoules (kJ)	Fat (g)
Filling:			
Fresh ricotta cheese	750 g	4238	75
Caster sugar	200 g	3400	0
Double cream	125 g	2375	63
Eggs	240 g	1500	28
Sambuca (Italian liquor)	30 mL	420	10
Icing sugar	10 g	170	0
Pastry base:			
Plain flour	150 g	2360	2
Almond meal	60 g	1460	32



Icing sugar	50 g	850	0
Cold butter	125 g	3938	100
Egg	60 g	375	7

- a see table b 21 085 c see table d 317 g
4 a 1757 b 26 g
5 a 2636 b 39 g
6 a 280 g b 23 g c 2.71 g
7 a 113.1 cm³ b 1696.5 cm³

10 Crazy cooking capers

- 1 No
2 a Circle b Yes
3 a 27.42 L b Yes c 58.92 L
4 30.54 cm 5 1713 mm 6 70 686 mm²
7 a 25 cm b 6.25 cm
8 a 30 cm b 1500 cm²
9 a 960 cm³ b 12 742.3 cm³
10 678.6 cm³ 11 2945.24 cm³
12 a 1.5 cm b 225 cm³
13 Yes, should only be 1.4953 L 14 2827.43 cm³
15 8.48 cm³ 16 398 cm²
17 a 314 cm² b 691 cm²
18

Radius of spoon	Volume of sphere	Volume of hemisphere (cubic centimetres)
0.7816	2	1
0.9847	4	2
1.3365	10	5
2.1216	40	20

11 Waste or waist?

- 1 \$443.04 2 8.395 3 2%
4 a 147 kg b \$245.10 c 2.8 kg d \$4.71 e (i) 49 (ii) 245
5 a \$US1000 b (i) \$US57.5 billion (thousand million) (ii) \$US115 billion (thousand million)
6 Between £UK1500 and £UK2400
7 a 0.11 kg b UK£21.350 billion
8 between 30 and 70 million tonnes
9 a (i) 7 700 000 tonnes (ii) 3 850 000 tonnes b 550 000
10 a 2 million m³ b 24 kW c 1000 d \$350 000

12 Paddock to plate!

- 1 a \$350 000 b \$656 250
2 a \$65 000 b 175 000 kilograms c \$64 925 d \$8 700 e \$56 225
3 \$1 137 500 4 0°Celsius
5 a 75 000 kilograms b 500 000 apples c 330 kilojoules d 63 750 kilograms
6 a 100 apples b 432 kg c 2700 apples d i 4 ii 1728 kg iii 10 800 e 864 pies
7 a \$4.00 b 60 cents c \$6912.00 d \$9072.00 e \$15 984 f \$9.25 g \$18.50

13 Cooking more or less!

- 1 No, only enough meat for 10 players 2 1 : 6
3 400 mL 4 200°C 5 2 to 32
6 350 to 360°F 7 30–33 ¾ min 8 42–45 min
9 a ¾ teaspoon b 1/12 to 1/8 teaspoon
10 a-c

Ingredient	Amount in recipe	Formula percentage	Double
Flour	225	39.5	450
Butter	227	39.9	454
Sour cream	112	19.7	224
Salt	5	0.9	10
Total	569.0	100.0	1138.0

- d 2250 g; 2270 g; 1120 g
11 a-c

Ingredient	Amount in recipe	Baker's percentage	Double
Flour	225	100.0	450
Butter	227	100.9	454
Sour cream	112	49.8	224
Salt	5	2.2	10
Total	569.0	252.9	1138.0

- d 569 g; 252.9%; $\frac{1 \text{ kg}}{252.9\%} = 395.4 \text{ g flour};$
100.9% = 398.96 g; 49.8% = 224.9 g; 2.2 % = 8.7 g

14 BMI - is that some sort of bike?

- 1 a 31 b 22 c See table below.

Volunteer	Height (cm)	Weight (kg)	Metric BMI
Karina Kalorie	175	95	31
David Deltoid	185	75	22
Jimmy Waylo	175	55	18
Patricia Pearsome	180	88	27

- 2 a

BMI standard	Weight status	Volunteer	BMI
Below 18.5	Way too thin	Jimmy Waylo	18
18.5–24.9	Way to go!	David Deltoid	22
25.0–29.9	Clothes way too tight	Patricia Pearsome	27.2
30+	Way too horizontal!	Karina Kalorie	31

- b BMI 22, David Deltoid
3 a 16 cm less b 5 cm more c 8 weeks
4

Volunteer	Age	Height (cm)	Weight (kg)	BEE	Gender
1 Karina Kalorie	19	175	95	7561	Female
2 David Deltoid	17	185	75	8037	Male
3 Jimmy Waylo	16	175	55	6695	Male
4 Patricia Pearsome	18	180	88	7341	Female

- 5 a 2.1 kg b 630 g c 46 200 kilojoules
6 a 50 g b ¼ cup of sugar



15 Nutrients in a lunch box

- 1 a 6 g b 33.6 g c 2000 µg d 0.8 mg e 0.35 mg
 2 19 g of dietary fibre, 12.4 g of protein, 0.3 mg of B2, 0.95 mg of B6
 3 a 0.9 µg b 75 mg c 360 µg d 900 mg e 7.6 mg
 4 Vitamin B12 5 5.4 µg, 1.65 mg 6 15.65 %
 7 a 15 mg b yes c 900 mg d 12.5 g e 75.5 g f 112.5 mg g 240 µg
 8 a 50.4 g b 19 g c 90 µg d 200 µg e 50.5 g
 9 a 833 mL b 2.2 cups

16 Global trends to overweight

- 1 a 4%; 17.5% b 13.5% c 17%, 6% d 1% e 1987–1998
 2 a 50 000 000 b 17 250 000 c \$2319
 3 a 1 500 000 b 6.98% c 96 d 22.5% e 750 000 f 11 000 000 g 9 100 000 h 3 458 000
 4 a 4 300 000 b 183 c 17 200 000 d 860 000 e 2 000 000

17 The nature of organics!

- 1 a 290 million b \$253 125 000 million c 15 billion
 2 a \$377 000 000 b \$490 100 000 c \$637 130 000 d \$828 269 000
 3 a 6.07 Ha b 4.047 Ha
 4 a 494 cows b 247 cows c 269 724 litres d \$124 073
 5 \$8.21 per kg
 6 a \$4125 b \$3125 c 24.24%
 7 a 593.03 acres b 240 million c 720 000 kg d 14 400 000 litres e \$216 000 000
 8 a 50 years b 1.49% c 121.5 mg d 176.2 mg e 156.8 mg

18 A sprouting good idea!

- 1 70 m² 2 28 m²
 3 a 2.9 m³ b 2.4 m³
 4 a 2 m³; 1 m³; 1 m³ b 4 m³; 2 m³; 2 m³ c 1.2 m³; 0.6 m³; 0.6 m³ d 2.43 m³
 5 40 m
 6

Plant	Sowing date	Time before picking (days)	Date to be picked
Tomato	1st November	135	16th March
Radish	1st October	30	31st October
Capsicum	1st November	135	16th March
Broccoli	25th October	135	9th March

- 7 a 16 b 64 c 144 d 256
 8 9
 9 48 (8 celery, 18 garlic, 12 lettuce, 10 tomatoes)

19 Okay, where's the fridge?

- 1 No answer given

2

a	5.5 m ²	10.3125 m ²	7.3125 m ²	5.5 m ²
b	0.5 m ²	0.5 m ²	0.5 m ²	0.5 m ²
c	0.5 m ²	0.5 m ²	0.5 m ²	0.5 m ²
d	0.5 m ²	0.5 m ²	0.75 m ²	0.75 m ²
e	1.25 m ²	1.5 m ²	1.5 m ²	1.5 m ²
f	0.9 m ³	0.9 m ³	1.35 m ³	1.35 m ³
g	2.75 m ²	7.3125 m ²	4.0625 m ²	2.25 m ²

- 3 a U-shaped and island b Galley and L-shaped

- 4 a U-shaped b Island
 5 1 m 6 1.25 m; 2.25 m; 2.574 m
 7 a 2.4 m b 1.7 m c 1.2 m

20 It's kool to be hot!

- 1 a 10 cups b 150 litres c 600 cups d-f See table.

Serves 10	Quantity for 600 serves	Cost for 10 serves	Cost for 600 serves
500 g peeled tomatoes	30 kg	\$0.60	\$36.00
1 kilo of potatoes	60 kg	\$1.20	\$72.00
250 g pumpkin	15 kg	\$0.50	\$30.00
2 onions	120	\$0.20	\$12.00
2 carrots	120	\$0.20	\$12.00
2 celery sticks	120	\$0.20	\$12.00
500 g cooked kidney beans	30 kg	\$1.50	\$90.00
1 zucchini	60	\$0.75	\$45.00
2 cloves of garlic	120	\$0.50	\$30.00
1 litre of chicken stock	60 L	\$2.00	\$120.00
¼ cup olive oil	15 cups	\$0.75	\$45.00
½ cup Italian parsley	30 cups	\$0.25	\$15.00
Total cost		8.65	\$519.00

- g \$900.00 h \$381 i \$1 714 500 j \$342 900
 2 a 30 000; 50 000; 350 000 b 10 c \$43 000
 d (i) $\frac{1}{120}$ (ii) $\frac{1}{12}$

21 This soup is way too kool!

- 1 a 1 050 000 b 367 500 c 14.7%
 2 a

Age group (years)	%	Population
<5	4	300 496
5–15	9	676 116
16–25	16	1 201 984
26–34	18	1 352 232
35–44	23	1 727 852
45–65	16	1 201 984
66+	14	1 051 736
Total	100	7 512 400

- b 3 080 084 c 41% d 1 293 635 e 17.22%
 f (i) 315 521 (ii) 105 174 g 1 609 155 h 21%
 i Yes, she is in the age group.

3 a

State	'000 people	% of population
New South Wales	6927.0	32.70%
Victoria	5246.1	24.77%
Queensland	4228.3	19.96%
South Australia	1591.9	7.52%
Western Australia	2130.8	10.06%
Tasmania	495.8	2.34%
Northern Territory	217.6	1.03%
Australian Capital Territory	343.1	1.62%
Total	21 180.6	100.00%

- b 8 472 240 c 3 812 508 d 18% e (i) 152 500.32 (ii) 190 625
 4 a 50% b 50% c 25%
 5 a 382 200 b 1 332 444 c 3 907 821
 6 a 551 250 b 1 940 453 c 4 765 635

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing **the highest quality learning outcomes with instant usability.**

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Maths in the Kitchen

The worksheets in this title feature the application of mathematical knowledge and techniques to various areas of cooking, health and food management. The worksheets are presented in the ongoing context of episodes of a TV cooking and lifestyle show hosted by celebrity chefs Ada and Lorenzo. Each topic is self-contained, and answers are provided to make marking quick and easy.

Topics include	
Order of operations Powers of 10 Multiplying and dividing fractions Converting measurements Cost and profit calculations Time calculations	Percentage and proportion Currency calculations Surface area, volume and mass Using formulae Geometry Interpreting graphs

Associated titles in the Instant Lessons series

- Maths In series
(Extreme sports, the Wild, Fashion, Science, Technology, Building design, Crime, Disasters, Food, Sport, the Community, the Environment, the Workplace, Travel)
 - Literacy in Maths (Books 1, 2; Number and Algebra; Data, Measurement and Geometry; A–Z of Maths Terms and Ideas; Maths Across the Curriculum)
 - Instant Lessons in Geometry (Books 1, 2)
 - Assignments in Junior Measurement (Books 1, 2, 3, 4)
 - Assignments in Junior Algebra (Books 1, 2)
- Developing Numeracy series
 - Numbers and the Number System (Books 1, 2, 3)
 - Chance and Data (Books 1, 2, 3)
 - Calculations (Books 1, 2, 3)
 - Algebra (Books 1, 2, 3)
 - Measurement and Space (Books 1, 2, 3)

Visit the Blake Education website or email info@blake.com.au for more information about the Instant Lessons series, other Mathematics titles, the full list of contents for all titles and ordering advice.



Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

